INFOTACT SOLUTIONS
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PROJECT REPORT

CareFlow

Clinical Pathway Process Mining

Process-mining and predictive-analytics study of emergency department patient journeys

Prepared by

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Akshar Patel

Abstract

Emergency departments need to understand not only how long patients stay, but how patients move through the department and where time is lost. This project, CareFlow: Clinical Pathway Process Mining, applies process mining and data-science preprocessing to an emergency department event log of 1,000 visits and 6,964 timestamped events covering ten clinical activities. The work was carried out at Infotact Solutions by a team of four interns.

A leakage-free preprocessing pipeline performs a chronological 80/20 split, treats structurally missing durations with explicit indicator flags, benchmarks four imputation methods (Iterative Bayesian Ridge achieved the lowest NRMSE of 0.614), applies z-score and min-max scaling fitted on training data only, and delivers a 41-feature arrival-time store with four prediction targets; all 16 automated checks passed. Exploratory analysis shows a mean length of stay of 107.5 minutes, a door-to-doctor wait of about 15 minutes for critical (ESI 1 and 2) patients versus about 49 minutes for others, and a midnight-crossing risk confined to evening and night admissions.

Process mining reveals six pathway variants whose median duration rises from 58 to 175 minutes. The slowest transitions are Lab Collection to Lab Result (32.6 minutes) and Triage to Doctor Assessment (30.7 minutes). Rework occurs in 23.5% of visits, always with the recorded reason of missing intake paperwork, and adds roughly 28 to 30 minutes to the affected journeys. The findings are presented through Power BI and Streamlit dashboards and translated into seven recommendations. Because the data are synthetic, the results demonstrate a reusable method rather than conclusions about a specific hospital.

Keywords: process mining, emergency department, patient pathway, length of stay, loopback, imputation, data leakage, Power BI

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1. Introduction

1.1 Background

Hospital emergency departments (EDs) are among the most operationally complex environments in healthcare. Patients arrive unpredictably, differ widely in severity, and move through a sequence of administrative, clinical and diagnostic steps before they are discharged or admitted. Every handover between steps is a place where time can be lost: a patient may wait for a physician, for a radiology report, or for a laboratory result, and some patients are sent back to an earlier step because information was incomplete. These delays lengthen the time each patient spends in the department, increase crowding, and make staffing difficult to plan.

Traditional hospital reporting summarises this complexity into a handful of aggregate figures, such as the average length of stay. Such figures say how long patients stayed, but not what happened to them while they were there. **Process mining** closes this gap. It uses the timestamped records that hospital information systems already collect (the event log) to reconstruct the actual journey of every patient, to discover the pathways that are followed in practice, and to measure the time spent between consecutive steps.

CareFlow: Clinical Pathway Process Mining applies this idea to an emergency department event log. The project was carried out as a one-month analytics assignment at Infotact Solutions, Bengaluru, by a team of four interns working in the Data Analytics domain. This report documents the complete work: the data, the preprocessing pipeline, the exploratory and process-mining analyses, the dashboards, the findings and the recommendations that follow from them.

1.2 Problem Statement

The emergency department studied in this project treats patients through ten recorded activities, yet management has no consolidated view of (i) which pathways patients actually follow, (ii) where in those pathways time accumulates, (iii) how often patients are sent back for repeated assessment, and (iv) which patient or operational characteristics are associated with longer stays. Without this visibility, decisions about staffing, diagnostic capacity and triage rules rest on intuition rather than evidence.

The problem addressed by this project can therefore be stated as follows: *given an event log of 1,000 emergency department visits, reconstruct the patient pathways, quantify delays and rework, prepare a leakage-free dataset suitable for predictive modelling, and communicate the results through reports and interactive dashboards.*

1.3 Objectives

- Collect, clean and validate the clinical event-log dataset and confirm that every case is complete and chronologically consistent.
- Identify the distinct patient pathways (process variants) and measure how common each one is and how long it takes.
- Measure the elapsed time between consecutive activities and rank the transitions that may represent bottlenecks.
- Detect loopbacks (rework) and estimate the additional time they add to a patient journey.
- Build a machine-learning-ready feature store that is free of data leakage, with documented imputation and scaling.
- Explore the influence of patient acuity, age, arrival time, shift and attending physician on length of stay.
- Present the results through Power BI and Streamlit dashboards and translate them into actionable recommendations.

1.4 Scope of Work

The scope follows the project assignment letter issued by Infotact Solutions (reference ITS/DA/1176), which defines the objective as providing practical exposure to real-world data analytics workflows: collecting, cleaning, analysing and visualising data, generating actionable insights, creating interactive dashboards and reports, and documenting the work to industry standards. The deliverables listed in the letter are the source code, project documentation, this project report, a presentation, and model evaluation results where applicable.

The assignment was scheduled for one month, from 5 October 2026 to 5 November 2026, within a three-month internship. The study is limited to a single emergency department represented by a de-identified, synthetic event log; no real patient data were used.

1.5 Team and Responsibilities

The work was divided so that each member owned one layer of the analysis. The four code branches submitted for this report correspond to these roles, and the main repository README consolidates them.

Table 1.1: Team members and areas of contribution

Member	Role	Main contribution
Akshar Patel	Lead Analyst, Data Engineer	Preprocessing pipeline (careflow_impute_scale.py): chronological split, structural and iterative imputation, benchmark of four imputers, dual scaling, scaler parameters, model-ready feature store, 16-point validation suite; consolidated project README.
Sagar Janathe	Collaborator (Member 3)	Exploratory analysis notebooks (Day 3 to Day 16): schema audit, demographics, activity frequency, LOS timeline, trace mining, transition matrix, acuity correlation, mid-project audit, outliers, cohort segmentation, peak loads, delay drivers; Power BI dashboard files; stakeholder summary.
Prachi Bhilare	Collaborator	Process-mining scripts (Day 3 to Day 18): event-log validation, loopback, transition and pathway analysis, waiting-time and bottleneck ranking, recommendations, static charts and the Streamlit interactive dashboard.
Prashanth	Collaborator	Team member listed in the project README. No branch from this member was supplied for this report; the Day 18 notes mention a "Member 4" who integrated dashboard pages, but the files do not say who that is.

1.6 Organisation of the Report

Chapter 2 introduces the concepts used in the study. Chapter 3 describes the requirements, methodology, tools and system design. Chapter 4 describes the dataset. Chapter 5 covers data preprocessing and feature engineering. Chapter 6 presents the exploratory analysis, and Chapter 7 the process-mining results. Chapter 8 describes the dashboards. Chapter 9 consolidates the findings, scenarios and recommendations, Chapter 10 documents validation and reconciliation of results across team members, Chapter 11 discusses ethics, privacy and governance, Chapter 12 covers limitations and future work, and Chapter 13 concludes. The appendices contain the data dictionary, the daily log, code excerpts, the repository inventory, a glossary and an extended event-log excerpt.

1.7 Deliverables

The assignment letter lists five categories of deliverables. Table 1.2 shows how each is addressed in the repository and in this report.

Table 1.2: Deliverables and where they are addressed

Deliverable	How it is addressed
Source code	Preprocessing script (akshar branch), 14 notebooks (sagar branch), 18 process-mining scripts and the Streamlit application (prachi branch)
Project documentation	README files on every branch, 16 analysis documents in the prachi docs folder, stakeholder summary and dashboard notes
Project report	This document, which consolidates the four branches
Presentation	Summary tables and figures in Chapters 6 to 9 are designed to be reused as presentation material
Model evaluation results	Imputation benchmark (Chapter 5), validation suite (Chapter 5) and statistical tests (Chapter 6); predictive-model training is listed as future work

1.8 Assumptions and Constraints

- The event log is complete: every visit that appears in the log has a registration and a discharge event, and the timestamps are recorded in a single consistent time zone.
- Timestamps have minute resolution, so intervals shorter than one minute cannot be observed.
- Activity labels are consistent across the log; no spelling variants of the same activity were present.
- Each visit is independent of the others. Crowding effects, in which one patient delays another, are not modelled explicitly, although they would be visible indirectly through arrival-hour patterns.
- The one-month assignment window limited the work to descriptive and diagnostic analytics and to data preparation for predictive models; model training was left for future work.

2. Background and Key Concepts

2.1 Process Mining

Process mining is a family of techniques that connects data science with process management. Instead of relying on a manually drawn process model, it discovers the process from event data. Three activities are usually distinguished: *discovery* (building a model from the log), *conformance checking* (comparing the log with a reference model) and *enhancement* (using the log to improve an existing model). This project concentrates on discovery and performance analysis, which are the most informative when no formal reference pathway is available [1].

2.2 Event Logs and Directly-Follows Graphs

An event log is a table in which each row records that a particular activity happened for a particular case at a particular time. Three attributes are mandatory: a case identifier (here, one emergency department visit), an activity name, and a timestamp. Additional attributes, such as the attending physician or the patient acuity, enrich the analysis. Sorting the events of each case by time yields its *trace*, the ordered list of activities the patient went through.

A **directly-follows graph (DFG)** is the simplest process model. Its nodes are activities and a directed edge from activity A to activity B is drawn whenever B immediately follows A in at least one trace; the edge is annotated with the number of times this happened and, optionally, the average elapsed time. Because it makes no assumption about concurrency, a DFG is a faithful summary of what the log contains and is the starting point for pathway and bottleneck analysis. The open-source PM4Py library [2] provides discovery and visualisation functions for it.

2.3 Emergency Department Flow and the Emergency Severity Index

A typical visit begins with registration, followed by triage, in which a nurse assigns an acuity level, and then by a physician assessment. Depending on the clinical picture, the physician may order imaging (X-ray with a radiology report) or laboratory tests, after which the results are reviewed and the patient is discharged or admitted. The acuity level used in this dataset follows the five-level **Emergency Severity Index (ESI)** [3], in which level 1 denotes a patient requiring immediate life-saving intervention and level 5 a non-urgent presentation.

Table 2.1: Emergency Severity Index levels used in the dataset

ESI level	Category	Meaning
1	Resuscitation	Immediate life-saving intervention required
2	Emergent	High-risk situation, severe pain or distress; should not wait
3	Urgent	Stable but expected to need several resources (for example imaging and laboratory tests)
4	Less urgent	Expected to need one resource
5	Non-urgent	Expected to need no resources beyond examination

2.4 Performance Measures

- **Length of stay (LOS):** time from registration to discharge, measured in minutes.
- **Door-to-doctor time:** time from arrival (registration) to the first physician assessment.
- **Transition time:** elapsed time between two consecutive recorded activities. It includes both waiting and the activity itself, so it is an indicator, not proof, of waiting.
- **Loopback (rework):** a return to an earlier step of the pathway, such as a second triage or a doctor reassessment.
- **Boarding indicator:** whether a visit crosses midnight, which signals pressure on beds and overnight staffing.

2.5 Data Leakage and Chronological Validation

When a model is trained to predict an outcome such as length of stay, it must only use information that would be available at the moment of prediction. Using statistics computed from future patients, or features that are only known after the visit has progressed, produces optimistic results that fail in practice; this is called *data leakage*. In operational data, a random train/test split can leak temporal trends between the sets. The CareFlow pipeline therefore splits cases chronologically and fits every imputer and scaler on the training period only.

2.6 Missing Data: Structural versus Genuine

Two kinds of missing values occur in clinical records. *Genuine* missingness means that a value existed but was not recorded. *Structural* missingness means that the value cannot exist because the event never happened: a patient who did not have an X-ray has no X-ray turnaround time. Treating structural gaps as genuine and filling them with the mean would fabricate procedures that never took place. The pipeline handles the two kinds differently (Chapter 5).

2.7 Process Model Notations

Several notations can represent a discovered process. A **Petri net** is a formal model with places, transitions and tokens; it can express concurrency, choice and loops precisely, and is the basis for conformance checking, but it can be hard for non-specialists to read. **BPMN** (Business

Process Model and Notation) uses familiar swim-lane diagrams and is widely used by process owners. A **directly-follows graph** shows only what actually followed what, together with frequencies and times, and makes no claim about parallelism.

For an emergency department where management first wants to know what happens and how long each step takes, the DFG is the most transparent choice. It is the model that the Day 6 workflow-discovery script produces with PM4Py, and it is the model on which the bottleneck ranking and loopback analysis in this report are based. A more formal model, such as an inductive-miner Petri net, is listed among the future extensions in Chapter 12.

2.8 Missing-Data Mechanisms and Imputation Methods

Statisticians distinguish three mechanisms by which values go missing. Data are **missing completely at random (MCAR)** when the chance of a value being missing does not depend on any data; **missing at random (MAR)** when it depends only on other observed variables; and **missing not at random (MNAR)** when it depends on the missing value itself. Imputation methods that use other columns, such as regression-based methods, are valid under MAR but not under MNAR. In this dataset no genuine missing values were found, so the choice of method was made on the basis of a masking experiment (Section 5.3).

Table 2.2: Imputation methods compared in the benchmark

Method	Principle	Strength	Weakness
Mean	Replace a gap by the column mean	Simple, no fitting beyond one statistic	Ignores relationships and shrinks variance
Median	Replace a gap by the column median	Robust to skew and outliers	Ignores relationships between columns
KNN (k = 5)	Average the same column over the five most similar rows	Uses the structure of the data and makes no linear assumption	Sensitive to scaling; slower on large data
Iterative Bayesian Ridge	Model each column from the others in rounds, using a regularised linear model	Uses correlations between columns; regularisation stabilises the fit	Assumes mostly linear relations; needs several passes

The iterative method is related to the technique known as multivariate imputation by chained equations. Each column with gaps is regressed on all the others in turn, and the cycle is repeated until the imputed values stabilise. Using a Bayesian ridge regressor adds a penalty on large coefficients, which keeps the fit stable when columns are correlated, as the duration columns are here.

2.9 Scaling and Encoding

Different features are measured on very different scales: age in tens of years, acuity from one to five, and durations from zero to several hundred minutes. Algorithms that rely on distances

or gradients are dominated by the features with the largest numeric ranges unless the features are rescaled. The two rescaling methods used in this project are defined in Chapter 5.

Categorical features such as the attending physician, the shift and the weekday have no natural numeric order and are represented by **one-hot encoding**: each category becomes a separate 0/1 column. Circular features such as the hour of the day are represented by their sine and cosine, so that the end of the cycle is adjacent to its start.

2.10 Statistical Methods Used

Table 2.3: Statistical methods applied in the analysis

Method	Used for	Where
Descriptive statistics (mean, median, standard deviation, percentiles)	Summarising durations and counts	Chapters 4, 6 and 7
Interquartile range (IQR) rule	Outlier screening: values above Q3 + 1.5 × IQR	Section 6.7
Pearson correlation	Linear association between features and targets	Sections 6.8 and 6.9
Chi-square test of independence	Loopback versus physician and versus acuity	Sections 6.3 and 6.6
Kruskal–Wallis test	Length of stay across physicians and shifts without assuming normality	Sections 6.5 and 6.6
Normalised root mean squared error (NRMSE)	Imputation accuracy	Section 5.3

The **chi-square test** compares observed counts in a contingency table with the counts expected if the two variables were unrelated; a small p-value indicates that the association is unlikely to be due to chance. The **Kruskal–Wallis test** is a rank-based alternative to one-way analysis of variance, suited to the skewed duration data. As with any significance test, a small p-value shows that a difference is unlikely to be random but does not show that it is large or that it has a particular cause.

2.11 Evaluation Metrics for Downstream Models

The feature store has been prepared for three prediction tasks. Table 2.4 lists the metrics recommended for each, so that models trained in the future can be compared consistently.

Table 2.4: Recommended evaluation metrics

Task	Metrics	Reason
Length-of-stay regression	Mean absolute error, root mean squared error, R^2, median absolute error	Error in minutes is directly interpretable; the median error is robust to the right-skewed target
Loopback classification (23.5% positive)	Precision-recall AUC, balanced accuracy, recall at fixed precision, Brier score	Accuracy is misleading for an imbalanced label; calibration matters when probabilities guide action
Midnight-crossing classification (8.1% positive)	Precision-recall AUC, recall, F1 score	The label is rare and the cost of missing a boarding case is high

3. Methodology and Tools

3.1 Overall Approach

The project followed a five-stage analytical pipeline, from raw event data to decision-ready insights (Figure 3.1). The stages were executed as an agile sprint with one deliverable per day, committed to branch-specific folders of a shared GitHub repository.

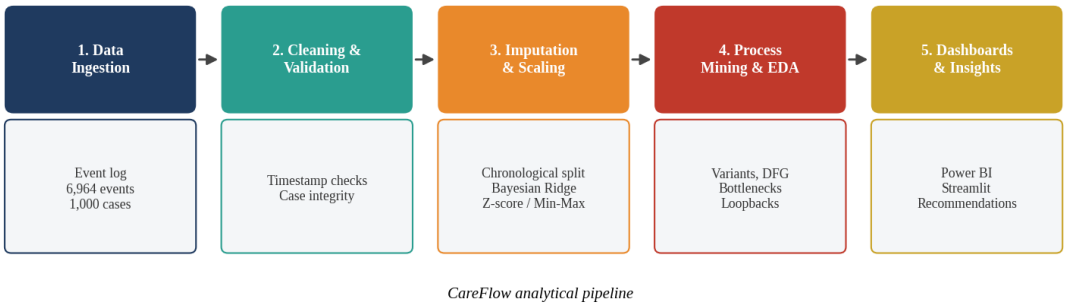


Figure 3.1: CareFlow analytical pipeline

3.2 Technology Stack

Table 3.1: Technologies used in the project

Layer	Tools	Purpose
Language and data handling	Python 3.10+ (tested on 3.11), Pandas, NumPy	Loading, cleaning and transforming event and case tables
Process mining	PM4Py	Event-log preparation, directly-follows graph and process-map discovery
Machine learning utilities	scikit-learn (IterativeImputer, KNNImputer, SimpleImputer, StandardScaler, MinMaxScaler, RandomForestRegressor) [4]	Imputation benchmark, scaling, feature importance
Visualisation	Matplotlib, Seaborn, Plotly, Streamlit, Power BI	Charts, interactive dashboard, four-page analytical report
Environment and version control	Jupyter Notebook, Git, GitHub (branches akshar, prachi, sagar, main)	Daily notebooks and collaborative development

3.3 Workstream 1: Preprocessing and Feature Store

The first workstream, owned by the lead analyst, produced the cleaned, imputed and scaled datasets used by everyone else. It was organised as a daily sprint whose milestones are listed in Table 3.2.

Table 3.2: Sprint milestones (Day 1 to Day 10) recorded in the main README

Day	Milestone	Outcome
1	Project kickoff	Environment set up (Python, PM4Py, Git); objectives, data-governance rules and roadmap defined
2	Preprocessing and validation	Raw event log ingested; timestamp chronology and case identifiers verified
3	Baseline EDA	Schema verification, profiling, master dataset consolidated
4	Demographics and acuity	170 critical-acuity cases (ESI 1 and 2) isolated
5	Activity frequency	Frequencies of all 10 activities; Triage the most frequent (1,235 events)
6	Duration and timeline	End-to-end durations and LOS by group
7	Trace mining	Six pathway variants; 705 loopback events in 235 cases
8	Transition matrix	Transition frequencies and heatmap
9	Acuity versus wait time	Correlation of acuity with durations
10	Mid-project audit	1,000 cases validated; audit summary exported

3.4 Workstream 2: Exploratory Analysis Notebooks

Table 3.3: Exploratory notebooks (Sagar Janathe)

Notebook	Topic	Output
day3_eda_baseline	Schema audit and baseline statistics	Verified 1,000 x 32 case table
day4_demographics_analysis	Age, acuity and department profiling	Demographic charts
day5_activity_frequency	Activity counts	Top-10 activity bar chart
day6_process_timeline	LOS statistics and spread	LOS summary and box plot
day7_trace_sequence_mining	Trace extraction and loopbacks	Common pathways; 235 loopback cases (23.5%)
day8_department_transition	Activity transition matrix	Heatmap of transition counts
day9_acuity_wait_time_correlation	Acuity, age and duration	Correlation matrix
day10_mid_project_performance_audit	Phase 2 checkpoint	day10_audit_summary.csv
day11_outlier_and_anomaly_detection	IQR outlier screening	careflow_cases_outliers.csv
day12_cohort_segmentation	High-acuity and standard cohorts	careflow_cases_segmented.csv
day13_time_series_peak_loads	Hourly load profile	day13_peak_loads_summary.csv
day14_feature_significance	Random-forest feature importance	day14_feature_significance_summary.csv
day15_manager_insights_dashboard	Executive summary view	day15_executive_dashboard_summary.csv
day16_final_metrics_summary	Consolidated metrics	careflow_final_metrics_summary.csv

3.5 Workstream 3: Process-Mining Scripts

Table 3.4: Process-mining scripts and documents (Prachi Bhilare)

Day	Script	Result
3	validate_event_log.py	Schema and timestamp checks on a 16-event sample log
4	loopback_analysis.py	705 loopback-marked events in 235 cases
5	activity_transition_analysis.py	5,964 transitions; 16 unique types
6	patient_workflow_discovery.py	PM4Py DFG and process map (SVG)
7	pathway_variation_analysis.py	Six unique pathways
8	patient_waiting_time_analysis.py	Elapsed time per transition
9	activity_frequency_analysis.py	Frequency of the 10 activities
10	activity_distribution_analysis.py	Events and patients per activity
11	patient_journey_sequence_analysis.py	Share of each complete sequence
12	patient_journey_transition_analysis.py	Transition shares
13	process_bottleneck_analysis.py	Transitions ranked by average elapsed time
14	workflow_performance_summary.py	Frequency and elapsed-time summary
15	workflow_improvement_recommendations.py	16 review recommendations
16	workflow_visualization_dashboard.py	Three static charts
17	careflow_interactive_dashboard.py	Streamlit dashboard
18	patient_journey_comparison.py	Pathway frequency versus duration

3.6 Repository Organisation

The repository is organised by branch. The main branch carries the consolidated README; the akshar branch carries the preprocessing script and all derived datasets; the sagar branch carries the notebooks, processed data folders (processed_old for baseline files and processed_new for machine-learning-ready files), and the Power BI dashboards; the prachi branch carries the src scripts, the processed summaries and the docs folder with one document per analysis day. Appendix D lists the contents of each branch.

3.7 System Requirements

Although CareFlow is an analytical study rather than a production software system, its deliverables were specified with explicit requirements so that the work could be reviewed and reproduced. The functional requirements describe what the pipeline and dashboards must do; the non-functional requirements describe how well they must do it.

Table 3.5: Functional requirements

ID	Requirement	Addressed by
FR1	Ingest the event log and case attributes and verify that they are complete	Preprocessing script; validate_event_log.py
FR2	Split cases chronologically into training and test sets	Preprocessing script, Section 5.1
FR3	Treat structural and genuine missing values differently	Preprocessing script, Section 5.2
FR4	Provide standardised and normalised versions of all numeric features	Preprocessing script, Section 5.4
FR5	Discover the process model and the distinct pathways	Process-mining scripts, Days 6, 7 and 11
FR6	Measure elapsed time between activities and rank transitions	Process-mining scripts, Days 8 and 13
FR7	Identify loopbacks and estimate their cost	Loopback analysis, Day 4; Section 7.6
FR8	Provide interactive dashboards with filters for acuity, shift, age group and weekend	Power BI pages; Streamlit application
FR9	Export all derived datasets as files for reuse	Processed data folders on all branches
FR10	Run an automated validation suite and log the results	scaling_checks.csv

Table 3.6: Non-functional requirements

ID	Requirement	How it is met
NFR1	Reproducibility: the same inputs must produce the same outputs	Fixed random seeds, saved scaler parameters, scripted pipeline
NFR2	No data leakage between training and test data	Chronological split; all statistics fitted on training rows only
NFR3	Privacy: no personal identifiers or confidential data	Case codes, physician codes, synthetic data
NFR4	Transparency: every transformation is documented	Daily documents, script comments, this report
NFR5	Usability: dashboards readable by non-technical managers	Four focused report pages, consistent layout and slicers
NFR6	Portability: runs on a standard Python installation	Python 3.10 or later with open-source libraries

3.8 System Architecture and Data Flow

Figure 3.2 shows the data flow of the system. The raw event log and case attributes enter the preprocessing stage, where they are validated, split and transformed. The resulting cleaned datasets feed three parallel consumers: the process-mining scripts, the exploratory notebooks and the feature store. The two dashboard layers read the derived summaries. The predictive-model stage is shown in a different colour because it is prepared for but not implemented in this assignment.

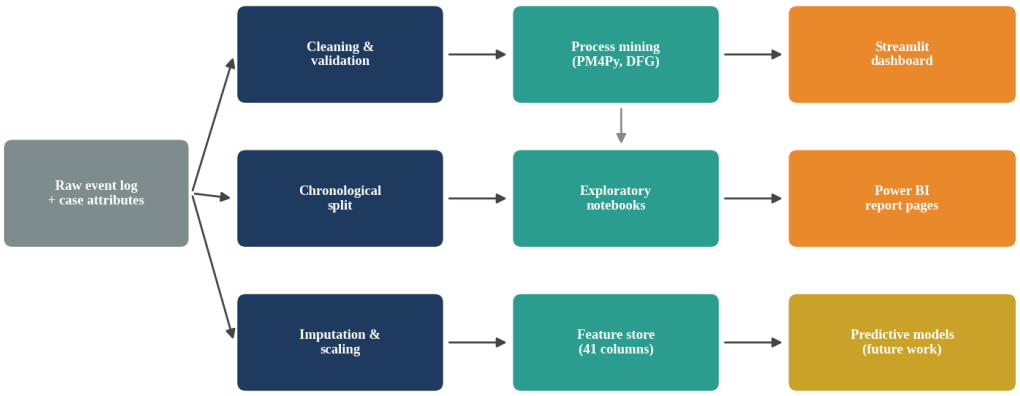


Figure 3.2: Data-flow view of the CareFlow system

Table 3.7: Modules of the system

Module	Inputs	Outputs	Owner
Preprocessing engine	Raw event log, case attributes	Imputed, standardised and normalised case tables; scaled event table; model-ready store; scaler parameters; checks	Akshar
Exploratory notebooks	Processed case and event tables	Statistical summaries, outlier and cohort files, charts	Sagar
Process-mining scripts	Event log	Transition, pathway, loopback and bottleneck summaries; process maps	Prachi
Power BI report	Imputed case table, scaled event table	Four report pages	Sagar
Streamlit application	Processed summary CSV files	Interactive workflow dashboard	Prachi

3.9 Environment Setup and Replication

The project can be reproduced on any machine with Python 3.10 or later (it was tested on Python 3.11) and the git version-control tool. The repository README documents the steps below.

Cloning the repository and creating an isolated environment

```
git clone https://github.com/akshar059/Infotact_P1_CareFlow.git
cd Infotact_P1_CareFlow

# Windows (PowerShell)
python -m venv venv
.\venv\Scripts\Activate.ps1

# Linux / macOS
python3 -m venv venv
source venv/bin/activate
```

Installing dependencies, running the pipeline and launching the notebooks

```
pip install pandas numpy scikit-learn pm4py matplotlib seaborn plotly jupyter

# automated imputation, scaling and validation
python data/processed_new/careflow_impute_scale.py

# exploratory notebooks
jupyter notebook notebooks/
```

Running the pipeline prints the 16 validation checks to the console and rewrites the derived datasets and the scaler-parameter file. Because all random operations use fixed seeds, repeated runs give identical results.

3.10 File Lineage

Table 3.8 traces each derived file back to the process that produces it and forward to its main consumers. This lineage is useful when the pipeline is rerun, because it shows which downstream files need to be refreshed.

Table 3.8: Lineage of the principal derived files

File	Produced by	Used by
careflow_cases_imputed.csv	Preprocessing engine	Notebooks, Power BI report, feature store
careflow_cases_standardized.csv, careflow_cases_normalized.csv	Preprocessing engine	Future model training
careflow_events_scaled.csv	Preprocessing engine	Process-mining scripts, Power BI report
careflow_model_ready_scaled.csv	Preprocessing engine	Future model training; correlation analysis
scaler_parameters.json	Preprocessing engine	Production scoring of new admissions
imputation_benchmark.csv, scaling_checks.csv	Preprocessing engine	Quality review; Chapter 5
workflow_performance_summary.csv and other summaries	Process-mining scripts	Streamlit application; recommendations
careflow_cases_outliers.csv, careflow_cases_segmented.csv	Day 11 and Day 12 notebooks	Cohort analysis; Power BI report

3.11 Quality Assurance Approach

Quality was managed at four levels. First, **input validation**: every analysis script checks that the required columns exist and that timestamps are valid before computing anything. Second, **automated checks** in the preprocessing engine verify the statistical properties of the transformed data. Third, **cross-branch consistency**: because three people analysed the same log independently, their results could be compared (Section 10.2). Fourth, **review**: the Day 10 mid-project audit validated the full set of 1,000 cases, and each branch kept a daily record of what was done in its documents.

3.12 Risk Register

Table 3.9: Project risks and mitigations

Risk	Impact	Mitigation
Data leakage inflates later model performance	Misleading accuracy; poor real-world results	Chronological split; statistics fitted on training only; saved parameters
Structural gaps filled with invented values	Fabricated procedures and durations	Explicit indicator flags; zero-fill only after verifying the flag
Over-interpretation of transition times as waiting	Wrong operational conclusions	Documented caution; recommendation to collect queue timestamps
Inconsistent figures between team members	Loss of credibility	Cross-checks and reconciliation (Chapter 10)
Synthetic data mistaken for real findings	Unjustified decisions	Explicit labelling on dashboards and in this report
Individual physicians judged on small samples	Unfair conclusions	Statistical testing and an explicit caution (Section 6.6)

4. Dataset Description

4.1 Source and Nature of the Data

The study uses an emergency department event log of 1,000 patient visits (cases) and 6,964 timestamped events. Case start times run from 1 January 2026 09:13 to 2 March 2026 06:58. The dashboards label the data as synthetic, and the baseline file is named careflow_synthetic_event_log.csv; this is consistent with the project's data-privacy rule that only public, de-identified or synthetic data may be used and that confidential patient information must not be published. All findings in this report therefore describe the supplied dataset and should be read as a demonstration of the method rather than as statements about a real hospital.

The data cover a single department (Emergency Department) and twelve attending physicians, identified only by codes DR01 to DR12. Patient age ranges from 18 to 90 years (mean 55.0). Visits are grouped into three age bands (18 to 39, 40 to 59 and 60 and above) and three eight-hour shifts (Day 07:00 to 15:00, Evening 15:00 to 23:00, Night 23:00 to 07:00).

4.2 Event-Level Table

The event table (careflow_events_scaled.csv, 6,964 rows and 27 columns) contains one row per activity. Its principal fields are listed below; the final eight columns are the standardised (_z) and min-max normalised (_mm) versions added by the preprocessing pipeline.

Table 4.1: Principal columns of the event table

Column	Description
case_id	Unique visit identifier (for example CF00001)
activity_name, activity_category	Activity label and its functional domain (Administrative, Clinical, Diagnostic)
timestamp	Time the activity was recorded, minute resolution
activity_order, activity_occurrence	Position of the event in the trace and repetition counter for the same activity
is_repeat_activity, is_loopback_event	Flags for repeated activities and for events on a loopback path
minutes_since_prev, minutes_since_case_start	Elapsed time since the previous event and since registration
resource, age, age_group, acuity, department	Attending physician code, patient age, age band, ESI level and department
loopback_reason, hour, day_of_week, split	Recorded reason for rework, hour of day, weekday, train/test assignment

4.3 Case-Level Table

The case table (careflow_cases_imputed.csv, 1,000 rows and 32 columns) has one row per visit and combines the case attributes with derived duration measures such as

registration_to_triage_min, door_to_doctor_min, xray_to_result_min, lab_turnaround_min, assessment_to_discharge_min and loopback_delay_min, together with the indicator flags has_xray, has_lab and has_reassessment. The full data dictionary is given in Appendix A.

4.4 Activity Catalogue

Table 4.2: The ten recorded activities

#	Activity	Category	Patients	% of patients	Events	Mean wait before (min)
1	Registration	Administrative	1,000	100.0	1,000	0.00
2	Triage	Administrative	1,000	100.0	1,235	11.87
3	Doctor Assessment	Clinical	1,000	100.0	1,000	30.74
4	X-Ray	Diagnostic	584	58.4	584	15.09
5	Radiology Result	Diagnostic	584	58.4	584	22.42
6	Lab Collection	Diagnostic	371	37.1	371	12.26
7	Lab Result	Diagnostic	371	37.1	371	32.56
8	Documentation Review	Administrative	584	58.4	584	9.88
9	Doctor Reassessment	Clinical	235	23.5	235	11.64
10	Discharge	Administrative	1,000	100.0	1,000	15.06

Registration, Doctor Assessment and Discharge occur in every visit. Triage has 1,235 events because 235 patients were triaged a second time. X-Ray, Radiology Result and Documentation Review each occur in 584 visits, and Lab Collection and Lab Result in 371 visits. Doctor Reassessment occurs in 235 visits. The mean wait before an activity is the average of minutes_since_prev for that activity.

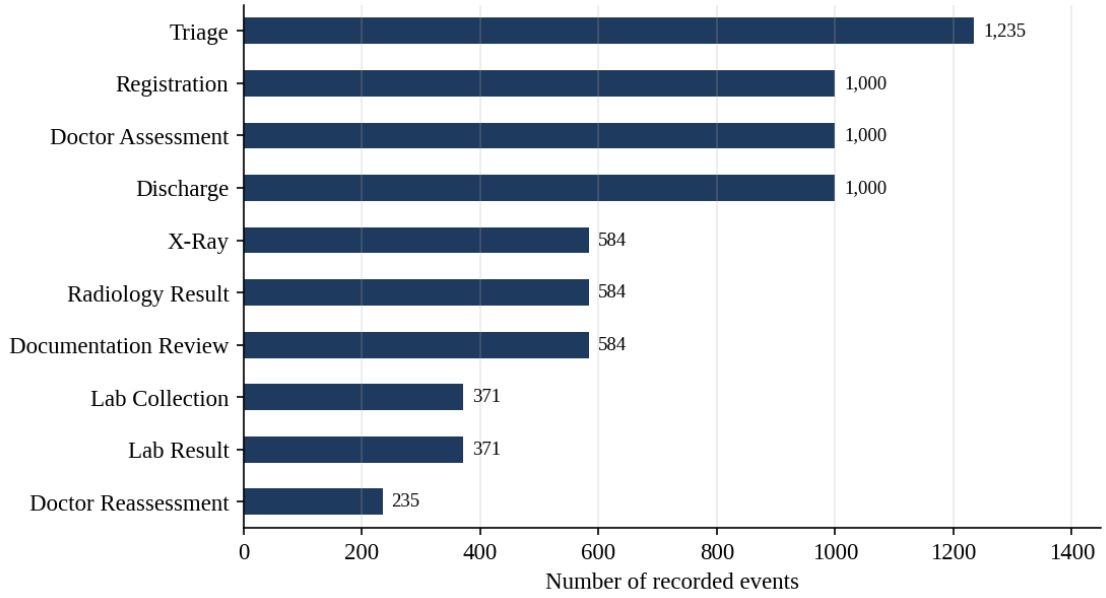


Figure 4.1: Number of recorded events per activity

4.5 Data Governance

Consistent with the project overview, no confidential patient information is included, personal identifiers are absent (patients are represented by case codes and physicians by DR codes), and the raw dataset was kept out of the Git commits of the process-mining branch. All analyses were performed on the supplied files without altering the source records.

4.6 Statistical Profile of the Numeric Attributes

Table 4.3 summarises the main numeric columns of the case table. The duration columns for X-ray, laboratory and loopback delay include zeros for visits where the procedure did not occur, which is why their medians and lower quartiles are zero.

Table 4.3: Descriptive statistics of the numeric case attributes

Attribute	Count	Mean	Std	Min	Q1	Median	Q3	Max
age	1000	55.0	21.7	18	36	55	75	90
acuity	1000	3.3	0.9	1	3	3	4	5
total duration min	1000	107.5	43.5	12	76	105	138	234
n events	1000	7.0	2.3	4	4	7	9	11
registration to triage min	1000	12.6	6.8	1	7	12	19	25
door to doctor min	1000	43.4	19.2	4	28	44	59	84
xray to result min	1000	13.1	12.5	0	0	13	24	35
lab turnaround min	1000	12.1	17.0	0	0	0	26	50
assessment to discharge min	1000	64.1	38.9	5	24	64	93	176
loopback delay min	1000	8.3	15.5	0	0	0	0	57

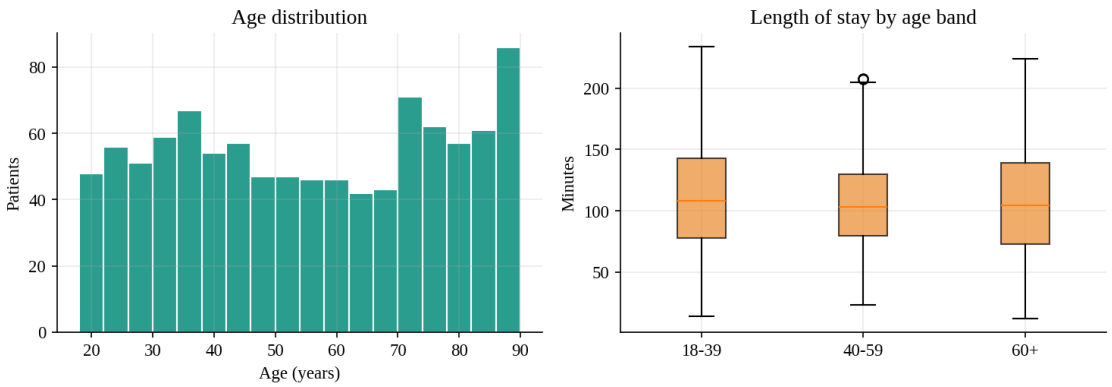


Figure 4.2: Age distribution and length of stay by age band

Age is spread across the adult range from 18 to 90 years. The middle half of patients are between 35.75 and 75 years old, with a median of 55 years, and one patient in ten is older than 85 years. The box plots show that the distribution of length of stay is similar in the three age bands, which agrees with the near-zero correlation between age and length of stay.

4.7 Activity-Level Timing Profile

For each activity, the table below gives the time that elapsed between the previous event and this one (the "wait before" measure), together with the time since registration at which the activity typically occurs. Registration has no previous event and therefore a wait of zero by definition. Triage combines first and repeat triages, which is why its figures are lower than those of a first triage alone.

Table 4.4: Timing profile of the ten activities (minutes)

Activity	Events	Mean wait	Median	P75	P90	Max	Mean time since start	Median time since start
Registration	1,000	0.0	0.0	0.0	0.0	0	0.0	0.0
Triage	1,235	11.9	11.0	17.0	22.0	25	27.4	15.0
Doctor Assessment	1,000	30.7	29.0	44.0	55.0	60	43.4	44.0
X-Ray	584	15.1	15.0	20.0	24.0	25	58.6	60.0
Radiology Result	584	22.4	22.0	29.0	33.0	35	81.0	80.0
Lab Collection	371	12.3	12.0	16.0	19.0	20	88.3	87.0
Lab Result	371	32.6	32.0	41.5	48.0	50	120.9	120.0
Documentation Review	584	9.9	8.0	12.0	20.0	25	94.4	95.0
Doctor Reassessment	235	11.6	12.0	16.0	18.0	20	116.6	118.0
Discharge	1,000	15.1	15.0	20.0	23.0	25	107.5	105.0

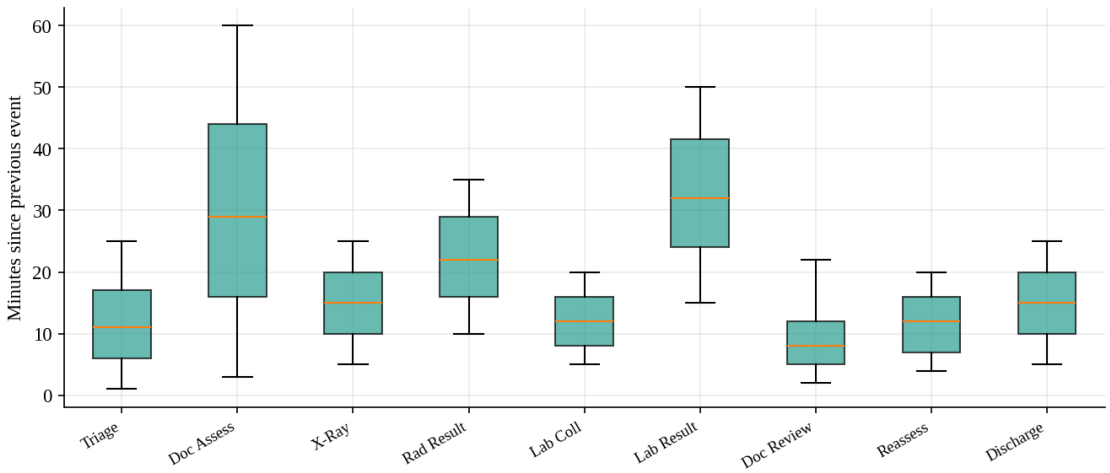


Figure 4.3: Distribution of the wait before each activity

Registration opens every visit and defines time zero. **Triage** follows registration, with a mean interval from the previous event of 11.9 minutes (median 11.0, 90th percentile 22.0); it occurs 1,235 times because 235 visits are triaged twice. **Doctor Assessment** is the longest first-contact step: it follows triage after a mean of 30.7 minutes (median 29.0), and the 90th percentile of 55.0 minutes shows that one visit in ten waits close to an hour. No wait exceeds 60 minutes, which suggests a cap in the way the data were generated.

X-Ray follows the assessment after a mean of 15.1 minutes, and **Radiology Result** follows X-Ray after a mean of 22.4 minutes (90th percentile 33.0). In the laboratory branch, **Lab Collection** follows the previous step after 12.3 minutes on average, and **Lab Result** follows collection after 32.6 minutes, the longest diagnostic interval. **Documentation Review** is a quick step (mean 9.9 minutes) that follows imaging in the rework path and in the review path. **Doctor Reassessment** follows after 11.6 minutes, and **Discharge** closes the visit after a mean of 15.1 minutes from the preceding event.

The last two columns of the table place each activity in time. A typical visit has its doctor assessment about 43 minutes after registration; for patients who need diagnostics, the radiology result arrives on average 81 minutes and the laboratory result 121 minutes after registration. Doctor Reassessment occurs on average 117 minutes after registration, which is why a loopback adds time late in the journey, after resources have already been spent on diagnostics.

4.8 Temporal Coverage

The 1,000 visits start on 61 calendar days between 1 January and 2 March 2026. On average 16.39 visits start per day (standard deviation 3.97); the busiest day was 2026-01-17 with 25 visits, and the quietest was 2026-03-02 with 5. The last day is only partly covered, because the final visit started in the early morning of 2 March. Figure 4.4 shows the daily series with a seven-day moving average, which stays close to its mean throughout the period, so the data show no strong long-term trend or seasonality.

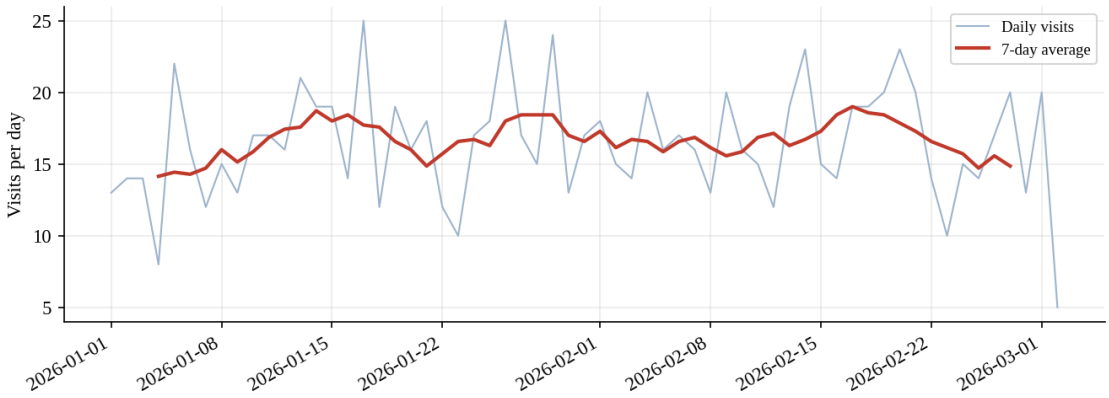


Figure 4.4: Visits starting per day with seven-day moving average

4.9 Example Event Trace

Table 4.5 shows the first sixteen rows of the event table. Each visit appears as a block of consecutive rows ordered by activity_order; minutes_since_prev is the interval to the previous row of the same visit.

Table 4.5: Excerpt of the event table

Case	Activity	Timestamp	Order	Min since prev	Physician	ESI
CF00001	Registration	2026-01-06 15:31	1	0	DR06	3
CF00001	Triage	2026-01-06 15:53	2	22	DR06	3
CF00001	Doctor Assessment	2026-01-06 16:13	3	20	DR06	3
CF00001	Discharge	2026-01-06 16:19	4	6	DR06	3
CF00002	Registration	2026-02-13 08:10	1	0	DR11	2
CF00002	Triage	2026-02-13 08:14	2	4	DR11	2
CF00002	Doctor Assessment	2026-02-13 08:20	3	6	DR11	2
CF00002	Discharge	2026-02-13 08:44	4	24	DR11	2
CF00003	Registration	2026-02-03 00:25	1	0	DR02	3
CF00003	Triage	2026-02-03 00:41	2	16	DR02	3
CF00003	Doctor Assessment	2026-02-03 01:34	3	53	DR02	3
CF00003	Discharge	2026-02-03 01:56	4	22	DR02	3
CF00004	Registration	2026-02-12 07:45	1	0	DR06	5
CF00004	Triage	2026-02-12 08:08	2	23	DR06	5
CF00004	Doctor Assessment	2026-02-12 08:56	3	48	DR06	5
CF00004	X-Ray	2026-02-12 09:05	4	9	DR06	5

5. Data Preprocessing and Feature Engineering

This chapter documents the preprocessing engine, careflow_impute_scale.py, which converts the cleaned case and event tables into imputed, standardised, normalised and model-ready datasets. Its design rests on three principles: no information from the future may influence a transformation, structural gaps must never be filled with invented values, and every transformation must be reproducible in production.

5.1 Chronological Train/Test Split

Cases are sorted by case_start and the first 80% are assigned to the training set (800 cases) and the remaining 20% to the test set (200 cases). The split cutoff is 18 February 2026 11:00. The script verifies that the latest training start is not later than the earliest test start. Every imputer and scaler is fitted on the training rows only and is then applied to the test rows without refitting.

Table 5.1: Composition of the chronological split

Set	Cases	Share	Loopback rate	Mean LOS (min)
Train	800	80%	23.6%	107.00
Test	200	20%	23.0%	109.49

5.2 Handling Missing Values

5.2.1 Structural missingness

Three duration fields are empty by construction: xray_to_result_min when the patient had no X-ray, lab_turnaround_min when there was no laboratory test, and loopback_delay_min when there was no reassessment. The pipeline first confirms that each field is missing exactly when its indicator (has_xray, has_lab, has_reassessment) equals zero, then sets the duration to 0.0 minutes and keeps the indicator as an explicit feature. This preserves the truth that the procedure did not occur and prevents a mean or median from implying a procedure that never took place.

5.2.2 Genuine missingness

For observed columns, genuine gaps would be filled with an iterative imputer fitted on the training set. The audit found no genuine missing values in this dataset (check 5 of the validation suite reports found = 0), so this path was kept for future data with gaps and was evaluated by a masking benchmark.

5.3 Imputation Benchmark

To select an imputation method, 10% of the observed values in eight numeric columns of the training set were masked at random, imputed by four methods, and compared with the true

values using the normalised root mean squared error, NRMSE = RMSE / standard deviation of the true values. A value of 1.0 means the method is no better than predicting the column mean.

Table 5.2: NRMSE of the four imputation methods on masked training data (lower is better)

Column	Iterative Bayesian Ridge	KNN (k = 5)	Mean	Median
acuity	0.8579	0.9230	1.0378	1.0881
age	1.0398	1.0999	1.0404	1.0390
assessment_to_discharge_min	0.1664	0.3339	0.9225	0.9233
door_to_doctor_min	0.4315	0.6527	0.9453	0.9411
n_events	0.3796	0.4632	1.0531	1.0517
registration_to_triage_min	0.8070	0.8588	1.0119	1.0209
start_hour	1.0486	1.1314	1.0485	1.0450
total_duration_min	0.1800	0.3587	1.0244	1.0232
MEAN OVER COLUMNS	0.6139	0.7277	1.0105	1.0165

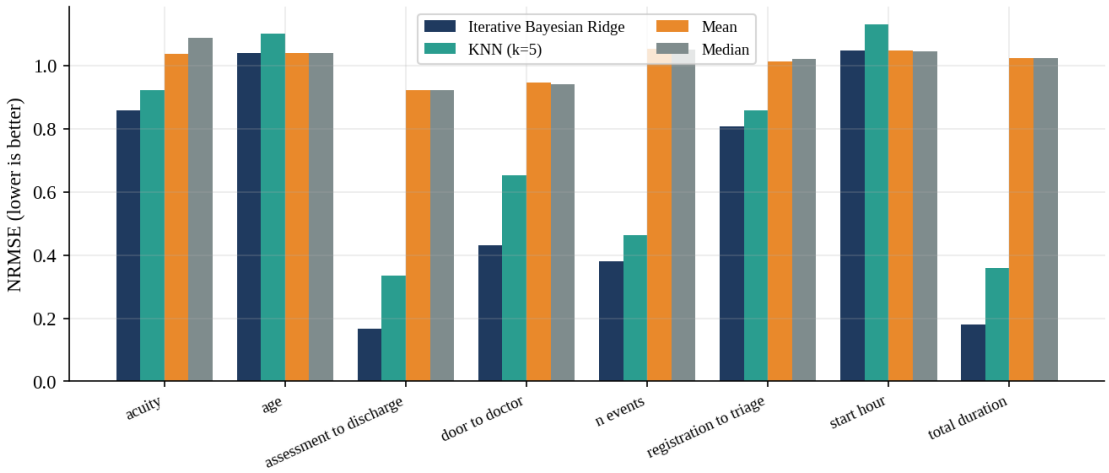


Figure 5.1: Imputation error by column and method

Iterative Bayesian Ridge achieved the lowest average NRMSE (0.614), which is 15.6% lower than KNN (0.728) and 39.2% lower than mean imputation (1.010). It performed best for the duration and event-count columns, which are strongly related to other variables. For age and start_hour, which are essentially independent of the other columns, no method beats the median or mean; the benchmark is therefore informative about where regression-based imputation helps and where it does not.

5.4 Scaling

Machine-learning algorithms have different needs: linear and gradient-based models benefit from zero-mean, unit-variance inputs, while distance-based models and neural networks often prefer bounded inputs. Two versions are therefore produced, both using training-set statistics:

- **Standardisation (z-score):** z = (x − μ_train) / σ_train. On the training set the mean is zero within 1.27×10^{-16} and the standard deviation is one (careflow_cases_standardized.csv).
- **Min-max normalisation:** x_norm = (x − min_train) / (max_train − min_train). Training values span exactly 0 to 1 (careflow_cases_normalized.csv). Test values outside the training range are deliberately not clipped; two such values were observed.
- **Event-level scaling:** minutes_since_prev (mean 15.36, SD 12.70, range 0 to 60) and minutes_since_case_start (mean 61.10, SD 48.59, range 0 to 234), together with age and acuity, are scaled across all 6,964 events (careflow_events_scaled.csv).

Table 5.3: Training-set parameters stored in scaler_parameters.json (case level)

Feature	Mean	Std. dev.	Min	Max
age	54.729	21.428	18	90
acuity	3.297	0.915	1	5
n_events	6.965	2.277	4	11
n_unique_activities	6.729	2.011	4	10
registration_to_triage_min	12.596	6.756	1	25
door_to_doctor_min	43.005	18.937	4	84
xray_to_result_min	12.963	12.387	0	35
lab_turnaround_min	12.054	16.966	0	50
assessment_to_discharge_min	63.994	39.409	5	176
loopback_delay_min	8.348	15.635	0	57
start_hour	11.550	6.918	0	23
total_duration_min	106.999	44.041	15	234

Persisting these constants in scaler_parameters.json allows a live admission to be transformed in production using the same constants, without database look-backs and without risk of leakage. The inverse transformation was verified to reconstruct the original values.

5.5 Cyclical Encoding of Arrival Time

Arrival hour is a circular variable: 23:00 and 00:00 are adjacent in time but 23 units apart on a linear scale. The hour is therefore projected onto the unit circle with start_hour_sin = sin(2π · hour / 24) and start_hour_cos = cos(2π · hour / 24). Euclidean distance between two arrival times then reflects true temporal proximity across midnight. These coordinates are already bounded in [−1, 1] and are not rescaled.

5.6 Model-Ready Feature Store

The file careflow_model_ready_scaled.csv (1,000 rows by 41 columns) contains only information available when the patient arrives, so that models built on it cannot look ahead. It is intended for length-of-stay prediction, loopback forecasting and bed-capacity planning.

Table 5.4: Composition of the model-ready feature store

Group	Columns	Encoding
Identifiers	case_id, case_start, split	Key, admission time, train/test label
Demographics	age, age_z, age_mm, age_group_18-39, 40-59, 60+	Raw, standardised, normalised and one-hot band
Severity	acuity, acuity_z, acuity_mm	ESI level raw, standardised and normalised
Arrival time	start_hour_sin, start_hour_cos	Cyclical coordinates
Calendar	is_weekend, seven start_weekday_* columns	Binary flag and one-hot weekday
Shift	shift_Day, shift_Evening, shift_Night	One-hot
Attending physician	resource_DR01 to resource_DR12	Twelve one-hot indicators
Targets	target_total_duration_min, target_total_duration_min_z, target_loopback, target_ends_next_day	Regression and classification labels

Four targets are provided. *target_total_duration_min* is the door-to-discharge length of stay in minutes and its standardised form (using training mean 106.999 and standard deviation 44.041). *target_loopback* is a binary label for visits that required rework (23.5% of cases). *target_ends_next_day* indicates that the visit crosses midnight (81 cases, 8.1%).

5.7 Automated Validation Suite

Each execution of the script runs 16 checks and logs them to scaling_checks.csv. All 16 passed in the submitted run (16 PASS, 0 FAIL).

Table 5.5: Results of the 16-point validation suite

#	Result	Check
1	PASS	split is chronological (max train start <= min test start)
2	PASS	xray_to_result_min is is NaN exactly when has_xray == 0
3	PASS	lab_turnaround_min is is NaN exactly when has_lab == 0
4	PASS	loopback_delay_min is is NaN exactly when has_reassessment == 0
5	PASS	genuine missing values in observed columns
6	PASS	no missing values remain after imputation
7	PASS	imputation benchmark completed
8	PASS	standardized train: mean ~ 0
9	PASS	standardized train: std ~ 1
10	PASS	normalized train within [0,1] and hits both ends
11	PASS	normalized test values outside [0,1] (expected possible, NOT clipped)
12	PASS	inverse transform recovers original values
13	PASS	event-level scaled rows == events rows
14	PASS	every event case has a split
15	PASS	model-ready features have no NaN
16	PASS	one-hot/sin/cos columns untouched (already bounded)

5.8 Guidelines for Downstream Modelling

- **LOS regression:** the distribution is right-skewed (maximum 234 minutes); robust losses such as Huber or absolute error are advisable, and the standardised target should be used for regularised linear models and neural networks.
- **Loopback classification:** with a 23.5% positive rate, accuracy is misleading; precision-recall AUC, balanced accuracy and the Brier score should be reported, with class weighting if needed.
- **Boarding forecasting:** arrival-hour harmonics and night-shift admission carry the strongest signal for crossing midnight (Section 6.8).
- **Reproducibility:** always apply the saved scaler parameters; never refit on new data.

5.9 Worked Example: One Visit Through the Pipeline

To make the transformations concrete, Table 5.6 follows a single visit, CF00278, from its raw attributes to its model-ready representation. The visit began at 09:13 on Thursday 1 January 2026, was assigned to physician DR09, and was a 22-year-old patient with ESI level 4. It is the earliest visit in the log and therefore belongs to the training set.

Table 5.6: Representation of visit CF00278 at each stage

Feature	Raw value	Standardised (z)	Min-max	Encoded form
age	22	−1.527	0.056	age_group_18-39 = 1
acuity	4	0.768	0.750	ordinal, 1 to 5
arrival hour	09:13	not applicable	not applicable	start_hour_sin = 0.707; start_hour_cos = -0.707
shift	Day	not applicable	not applicable	shift_Day = 1
weekday	Thursday	not applicable	not applicable	start_weekday_Thursday = 1
physician	DR09	not applicable	not applicable	resource_DR09 = 1
total duration (target)	135 min	0.636	not applicable	target_total_duration_min = 135
loopback (target)	yes	not applicable	not applicable	target_loopback = 1

The z-score of age follows from the training mean and standard deviation in Table 5.3: a 22-year-old patient is about 1.5 standard deviations younger than the average training patient. The min-max value of 0.056 shows that 22 years is close to the youngest end of the observed range of 18 to 90 years.

5.10 Train and Test Comparability

A chronological split can produce training and test sets that differ in character if the process changes over time. Figure 5.2 compares the length-of-stay distributions of the two sets. Their summary statistics are close: the training set has a mean of 107.0 minutes (standard deviation 44.1) and the test set a mean of 109.5 minutes (standard deviation 41.3), and the medians are

103.5 and 109.5 minutes. The loopback rates are also close (23.6% and 23.0%). The test set is therefore a fair representation of the training distribution, and a model trained on the first four-fifths of the period can be evaluated meaningfully on the last fifth.

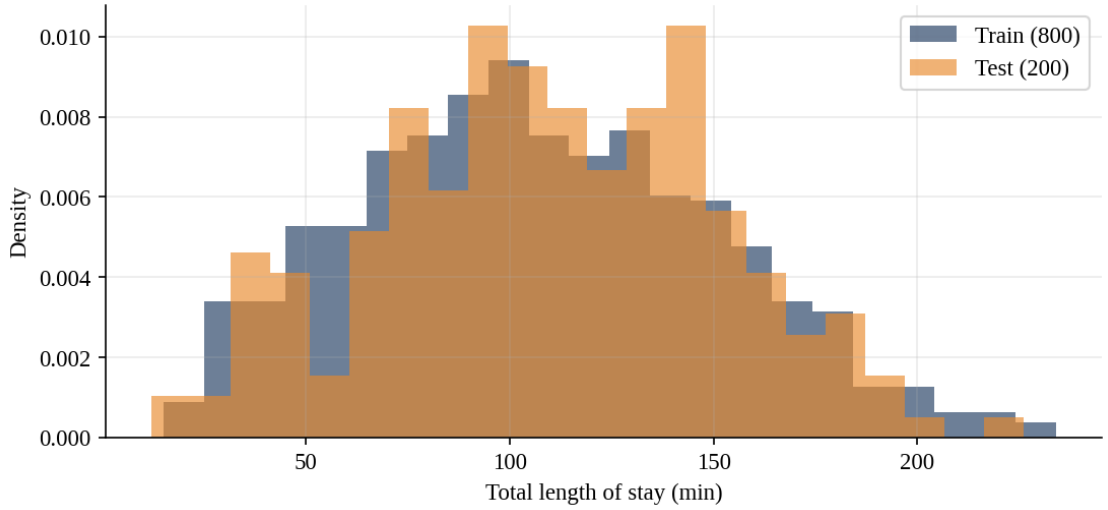


Figure 5.2: Length-of-stay distribution in the training and test sets

5.11 Effect of Scaling on a Feature

Scaling changes the units of a feature but not the shape of its distribution. Figure 5.3 shows the age distribution before and after each transformation: the three histograms have exactly the same shape; only the horizontal axis differs. Standardised ages are centred on zero with unit spread, and min-max ages lie between zero and one, with the training minimum at zero and the training maximum at one.

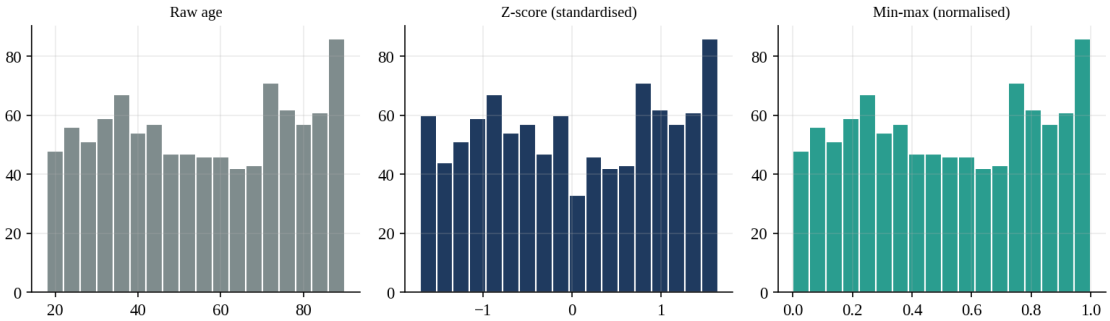


Figure 5.3: Age before scaling, after standardisation and after min-max normalisation

5.12 Cyclical Encoding in Practice

Figure 5.4 places the 24 hours of the day on the unit circle. Hour 23 and hour 0 are neighbours on the circle even though they are 23 units apart on a number line. Table 5.7 gives the coordinates for four representative hours; midnight has the largest cosine and noon the smallest, which is why the cosine of the arrival hour is the strongest correlate of crossing midnight.

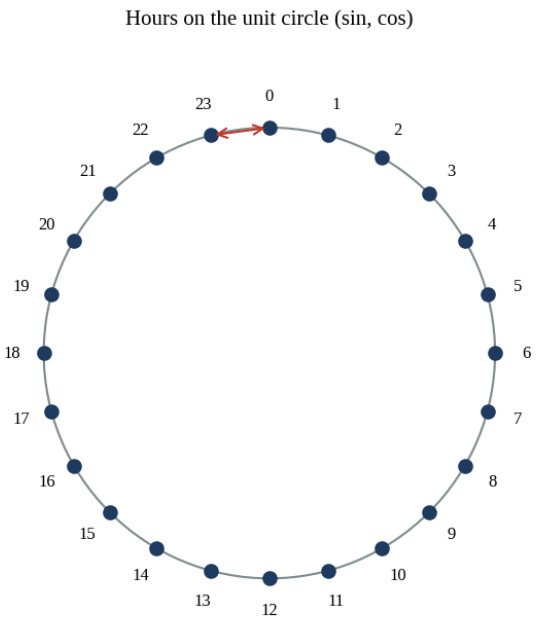


Figure 5.4: Hours of the day on the unit circle

Table 5.7: Cyclical coordinates of selected arrival hours

Hour	start_hour_sin	start_hour_cos	Interpretation
0	0.000	1.000	Midnight: cosine at its maximum
6	1.000	0.000	Early morning: sine at its maximum
12	0.000	-1.000	Noon: cosine at its minimum
18	-1.000	-0.000	Evening: sine at its minimum
23	-0.259	0.966	Close to midnight, adjacent to hour 0

5.13 Design Decisions and Alternatives

Table 5.8: Principal design decisions in the preprocessing engine

Decision	Alternative considered	Reason for the choice
Chronological 80/20 split	Random split	Prevents temporal leakage and mimics deployment, where the model predicts future visits
Zero-fill structural gaps with indicator flags	Mean or iterative imputation of every gap	The procedure never happened, so an imputed duration would be fictitious
Iterative Bayesian Ridge for genuine gaps	KNN, mean or median	Lowest average NRMSE in the masking benchmark
Both z-score and min-max versions	A single scaling	Different model families have different needs; both are cheap to store
No clipping of out-of-range test values	Clip to the training range	Preserves information and exposes drift; only two values were affected
Persist parameters in JSON	Recompute on demand	Guarantees identical transformation in production and avoids look-backs
Sine and cosine of the hour	Raw hour or hour bins	Preserves adjacency of late-night and early-morning hours
Arrival-time features only in the feature store	Include all duration measures	Duration measures are known only after the visit progresses and would leak the target

6. Exploratory Data Analysis

The exploratory analysis was carried out in the Day 3 to Day 16 notebooks and re-verified against the processed files for this report. It profiles patient characteristics, the length-of-stay distribution, temporal patterns and physician workload.

6.1 Overview of Key Performance Indicators

Table 6.1: Headline indicators of the dataset

Indicator	Value
Patient visits (cases)	1,000
Recorded events	6,964
Mean length of stay	107.5 min (median 105, SD 43.5)
Length-of-stay range	12 to 234 min (interquartile range 76 to 138.25)
Mean registration-to-triage time	12.6 min
Mean door-to-doctor time	43.4 min
Mean doctor-assessment-to-discharge time	64.1 min
Visits with a loopback	235 (23.5%)
Critical-acuity visits (ESI 1 or 2)	170 (17.0%)
Visits crossing midnight	81 (8.1%)

6.2 Patient Demographics

Patients are predominantly older: the 60-and-above band accounts for 44.6% of visits. Mean length of stay is highest for the 18 to 39 band (110.1 minutes) and lowest for the 40 to 59 band (106.0 minutes), a difference that is small relative to the standard deviation of 43.5 minutes. Age has almost no linear relationship with length of stay ($r = -0.042$).

Table 6.2: Visits by age band

Age band	Visits	Share (%)	Mean LOS (min)	Loopback rate (%)	Mean acuity
18-39	306	30.6	110.1	26.5	3.23
40-59	248	24.8	106.0	20.6	3.35
60+	446	44.6	106.5	23.1	3.31

6.3 Acuity and Triage Velocity

Acuity levels 3 and 4 dominate (42.9% and 30.9% of visits); only 2.6% of visits are ESI 1. The most striking result is a sharp split in door-to-doctor time between critical and non-critical patients: ESI 1 and 2 patients see a physician after roughly 14 to 16 minutes, while ESI 3 to 5 patients wait about 49 to 50 minutes, three times longer (Figure 6.1). The same split appears in

the registration-to-triage time (about 4 minutes versus 14 minutes). This shows that the department's priority rule works for the critical group.

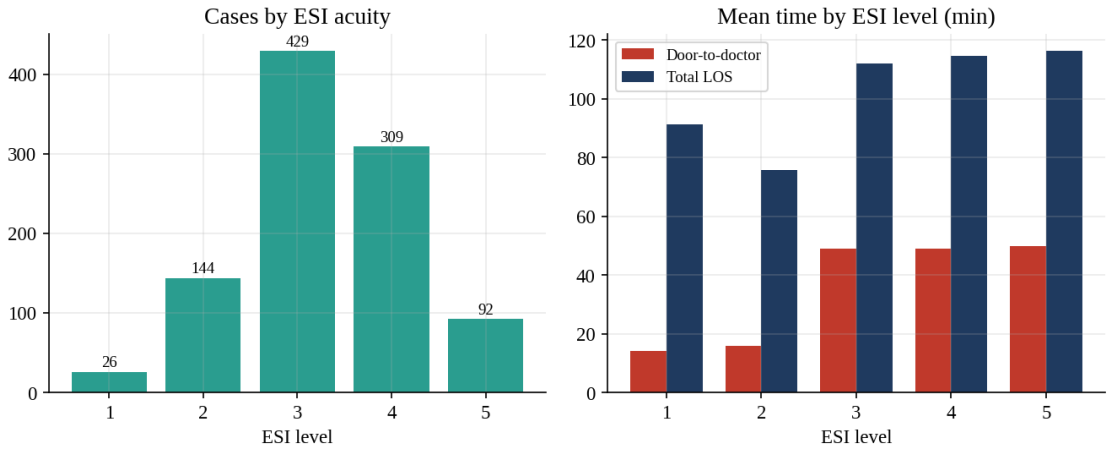


Figure 6.1: Visits by ESI level and mean door-to-doctor time and length of stay

Table 6.3: Triage velocity and length of stay by ESI level

ESI	Visits	Share (%)	Reg. to triage (min)	Door to doctor (min)	Assess. to discharge (min)	Mean LOS (min)	Loopback (%)
1	26	2.6	3.8	14.1	77.3	91.4	23.1
2	144	14.4	4.2	16.0	59.8	75.8	26.4
3	429	42.9	14.3	48.9	63.2	112.1	21.7
4	309	30.9	14.3	49.0	65.7	114.6	24.3
5	92	9.2	14.7	50.0	66.3	116.3	25.0

An important nuance is that mean length of stay is *shorter* for ESI 1 and 2 (91.4 and 75.8 minutes) than for ESI 3 to 5 (about 112 to 116 minutes). The positive correlation of acuity score with length of stay ($r = 0.232$) therefore means that less urgent patients stay longer in this dataset, largely because they wait longer before the first doctor contact. Loopback rate is not associated with acuity (chi-square test $p = 0.79$).

6.4 Length-of-Stay Distribution

Length of stay ranges from 12 to 234 minutes with mean 107.5 and median 105 minutes. The distribution is broad rather than sharply peaked, which reflects the mixture of six very different pathways (Section 7.5): direct visits finish in about an hour, while the most complex pathway averages almost three hours.

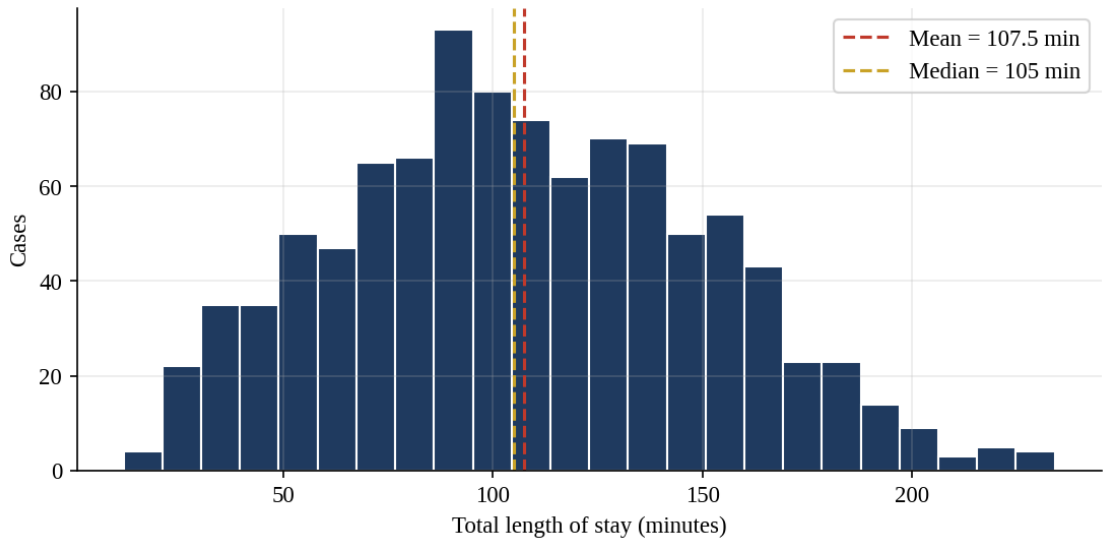


Figure 6.2: Distribution of total length of stay

6.5 Temporal Patterns

6.5.1 Hour of arrival

Arrivals are spread across the whole day with no pronounced peak. The busiest arrival hour is 16:00 (60 arrivals), followed by 15:00 (50), and the quietest is 12:00 (28). Sagar's Day 13 hourly profile counts *events* rather than arrivals, because each visit contributes between 4 and 11 events; event load is highest from 16:00 to 18:00 (332 to 349 events per hour), which agrees with the arrival peak in the late afternoon.

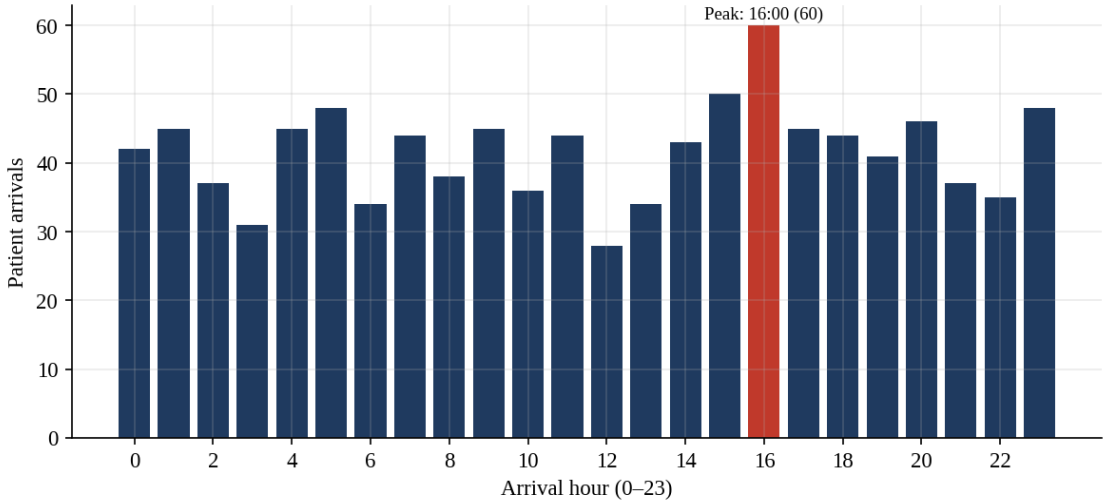


Figure 6.3: Patient arrivals by hour of day

6.5.2 Day of week and weekend

Saturday has the most visits (162) and Wednesday the fewest (132). Weekend visits (29.7%) have almost the same mean length of stay as weekday visits (107.9 versus 107.3 minutes).

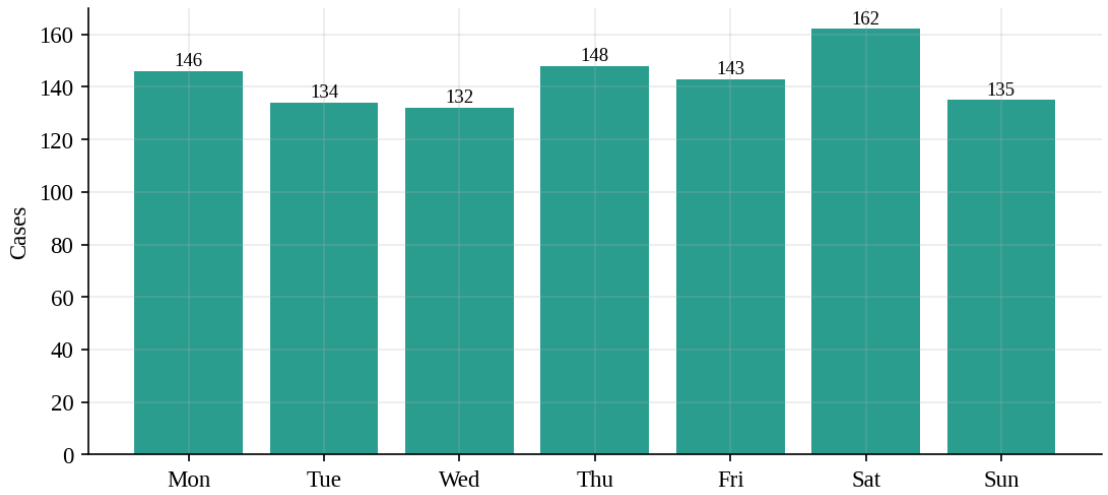


Figure 6.4: Visits by day of week

6.5.3 Shift

Evening is the busiest shift (358 visits), followed by Night (330) and Day (312). Mean length of stay differs little between shifts (Day 110.6, Evening 105.1, Night 107.1 minutes; Kruskal–Wallis p = 0.26). The shift effect is clearest for the boarding indicator: no day-shift patient crosses midnight, whereas 36 evening and 45 night admissions do.

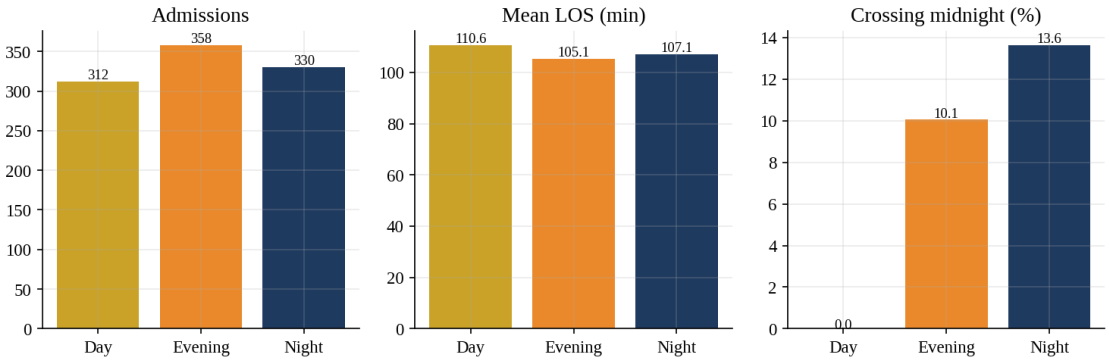


Figure 6.5: Admissions, mean length of stay and midnight crossing by shift

Table 6.4: Shift-level summary

Shift	Visits	Share (%)	Mean LOS (min)	Loopback (%)	Cross midnight (n)	Cross midnight (%)
Day	312	31.2	110.6	25.6	0	0.0
Evening	358	35.8	105.1	21.5	36	10.1
Night	330	33.0	107.1	23.6	45	13.6

6.6 Physician Workload

Case volume ranges from 64 visits (DR10) to 106 visits (DR07). Mean length of stay ranges from 101.0 minutes (DR01) to 116.4 minutes (DR12). Loopback rates vary more widely, from 12.3% (DR11) to 34.1% (DR12). A chi-square test of loopback against physician gives p = 0.025, indicating that the differences are unlikely to be pure chance, whereas the differences in

length of stay are not statistically significant (Kruskal–Wallis p = 0.31). With only 64 to 106 visits per physician, these comparisons should be used to prompt a review, not to judge individuals.

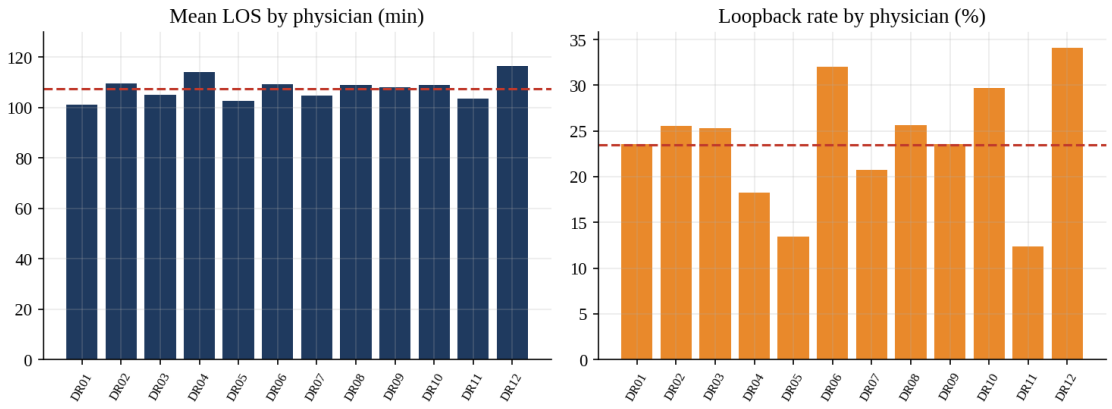


Figure 6.6: Mean length of stay and loopback rate by attending physician

Table 6.5: Workload and outcomes by attending physician

Physician	Visits	Share (%)	Mean LOS (min)	Loopback (%)	Mean acuity
DR01	102	10.2	101.0	23.5	3.23
DR02	86	8.6	109.6	25.6	3.35
DR03	79	7.9	105.2	25.3	3.23
DR04	82	8.2	114.1	18.3	3.32
DR05	89	8.9	102.5	13.5	3.28
DR06	75	7.5	109.2	32.0	3.32
DR07	106	10.6	104.7	20.8	3.37
DR08	74	7.4	109.0	25.7	3.31
DR09	85	8.5	108.1	23.5	3.25
DR10	64	6.4	108.9	29.7	3.34
DR11	73	7.3	103.6	12.3	3.26
DR12	85	8.5	116.4	34.1	3.32

6.7 Outlier and Cohort Analysis

Outliers were screened with the interquartile range rule (values above Q3 + 1.5 × IQR). For total duration, Q1 = 76, Q3 = 138.25 and the upper fence is 231.6 minutes; one visit (234 minutes) exceeds it and is flagged in the duration_outlier_flag column. The Day 11 notebook, which screened assessment_to_discharge_min, found no outliers (upper fence 197.1 minutes, maximum 176). The dataset is therefore remarkably free of extreme values, which is typical of synthetic data and means that long stays here are explained by pathway complexity rather than by anomalies.

The Day 12 notebook segmented visits into a high-acuity cohort (ESI 1 and 2, 170 visits) and a standard cohort (830 visits). The mean assessment-to-discharge time was 62.4 minutes in the

first group and 64.5 minutes in the second, so the two cohorts are almost equal once the doctor has seen the patient; the difference between them arises before the doctor contact (Section 6.3).

6.8 Drivers of Length of Stay

Pearson correlations over the model-ready features give a consistent ranking (Table 6.6). By far the strongest correlate of length of stay is whether the visit involved a loopback (r = 0.524), followed by acuity score (r = 0.232). Age (r = −0.042) and individual physician or shift indicators are weak. For the boarding indicator, the cosine of the arrival hour is the strongest predictor (r = 0.387), because the cosine peaks at midnight, followed by night-shift admission (r = 0.142).

Table 6.6: Selected correlations with the engineered targets

Target	Feature	Pearson r
target_total_duration_min	target_loopback	0.524
target_total_duration_min	acuity	0.232
target_total_duration_min	target_ends_next_day	0.123
target_total_duration_min	age	-0.042
target_ends_next_day	start_hour_cos	0.387
target_ends_next_day	shift_Night	0.142
target_ends_next_day	start_hour_sin	-0.146

Note on the Day 14 feature-importance notebook
The Day 14 random-forest model was fitted with a single predictor (acuity), so its reported importance of 1.0 is trivially true and says only that acuity was the only feature supplied. The correlation ranking above, computed over the full set of arrival-time features, gives the broader picture.

6.9 Correlation Structure

Figure 6.7 shows the full correlation matrix of the targets and the principal features. Three groups stand out. First, length of stay is related to loopback (0.52) and, more weakly, to acuity (0.23); all other features have negligible correlation with it. Second, the boarding indicator is related to the hour coordinates (cosine 0.39, sine −0.15) and to night-shift admission (0.14). Third, the hour coordinates are strongly correlated with the shift indicators (for example sine and evening shift, −0.83), which is expected because the shifts are defined by hour ranges. When both are used in a linear model, this redundancy should be handled with regularisation.

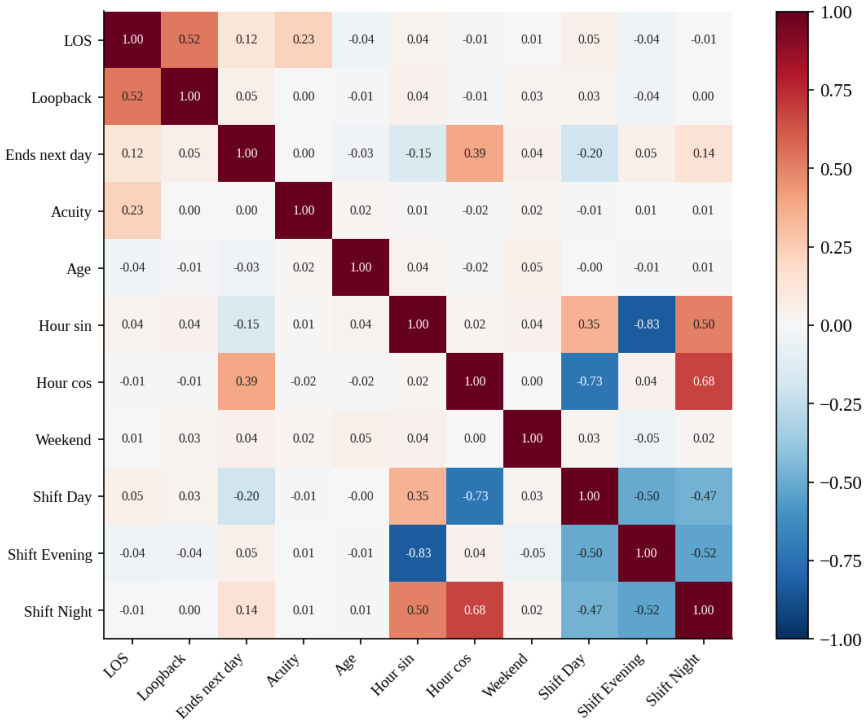


Figure 6.7: Pearson correlation matrix of targets and arrival-time features

6.10 Arrival Patterns by Weekday and Hour

Figure 6.8 combines the day of the week and the hour of arrival. The darker cells mark the busiest combinations. Because there are only 1,000 visits spread over 168 weekday-hour cells, individual cells contain between zero and ten visits and the map should be read for broad patterns rather than individual cells. The broad pattern is a mild concentration of arrivals in the afternoon and early evening and smaller peaks around 05:00 and 23:00.

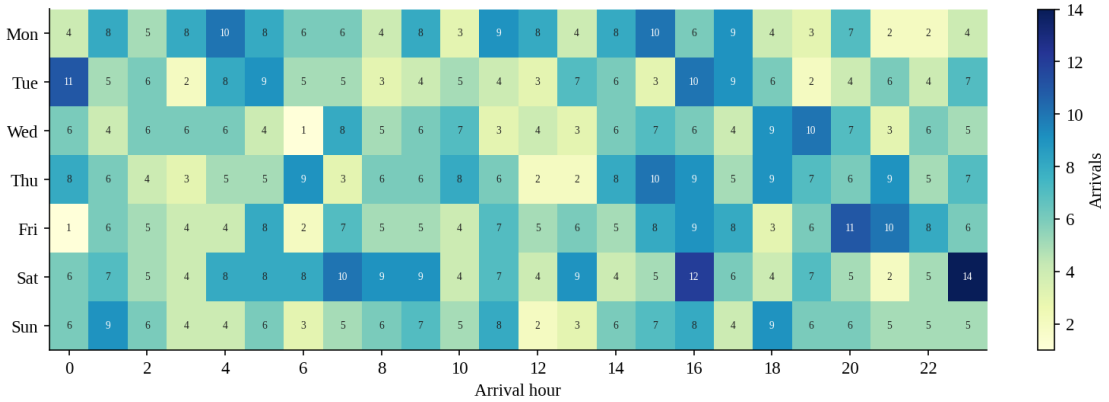


Figure 6.8: Arrivals by weekday and hour of day

6.11 Door-to-Doctor Time Across the Day

Figure 6.9 relates the arrival volume to the mean door-to-doctor time for each hour. If the department were congested at busy hours, the red line would rise where the bars are tall. In this dataset the line stays between about 38 and 49 minutes without following the arrival volume closely; the door-to-doctor time is determined mainly by the acuity mix of the arrivals in each

hour rather than by the volume. This is a further sign that the synthetic log was generated from acuity-dependent distributions and not from a queueing process, and it limits what can be concluded about crowding.

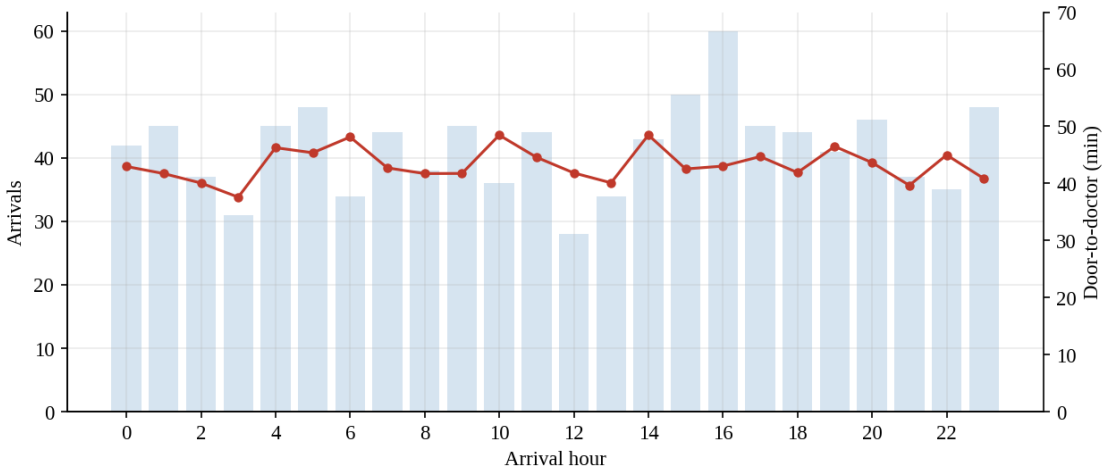


Figure 6.9: Arrivals and mean door-to-doctor time by hour

6.12 Acuity and Shift Interaction

Figure 6.10 shows the mean length of stay for each combination of acuity and shift. Acuity dominates: ESI 2 visits are shortest in every shift and ESI 3 to 5 visits are longest, whereas the differences between shifts within one acuity level are comparatively small. This supports the earlier finding that shift has little influence on the length of stay and that its main operational effect is on boarding past midnight.

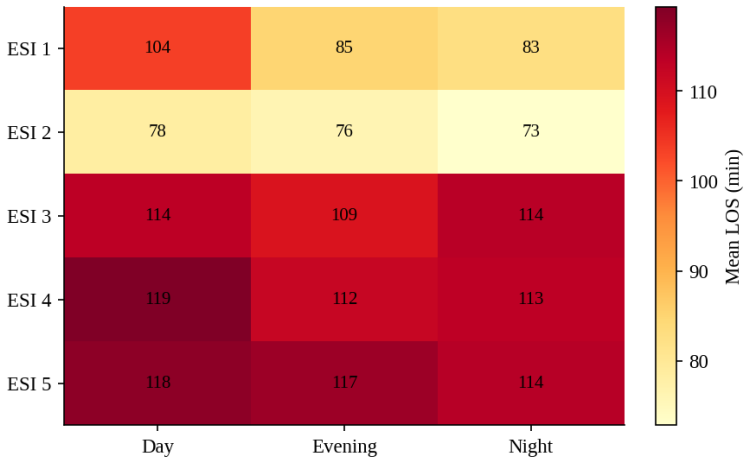


Figure 6.10: Mean length of stay by acuity and shift

6.13 Loopback Profile

Figure 6.11 shows that the loopback rate does not vary systematically with weekday or age band: the rates lie between 19% and 27% around the overall 23.5%, with no consistent pattern, and differences of this size are within the sampling variation expected for groups of 100 to 450 visits. Combined with the earlier finding that the rate does not depend on acuity ($p = 0.79$), this

indicates that rework is not caused by a particular type of patient or a particular day. The only factor that was found to differ significantly is the attending physician ($p = 0.025$), and the pathway: loopbacks occur only in the two pathways that include radiology.

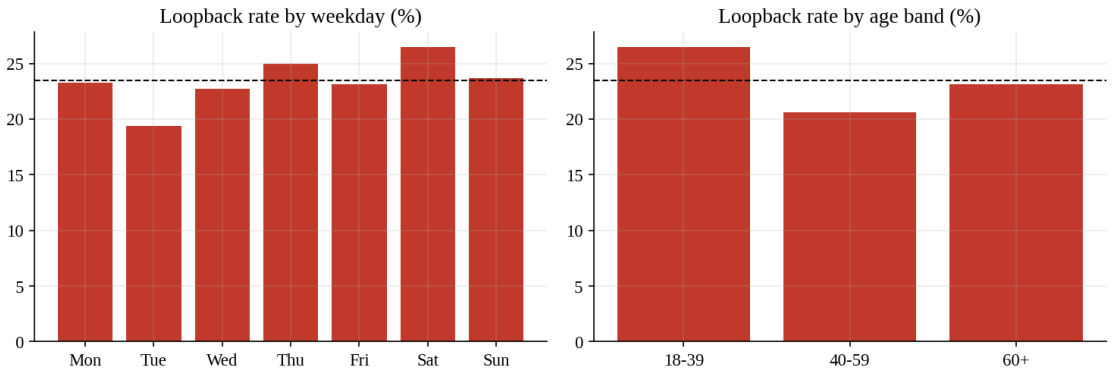


Figure 6.11: Loopback rate by weekday and by age band

6.14 Pathway Mix and Time Decomposition

Figure 6.12 shows how the six pathways are distributed across acuity levels. The share of each pathway varies modestly from one acuity level to another (for example, the direct pathway V1 accounts for 19% of ESI 1, 31% of ESI 2 and 25% to 27% of ESI 3 to 5 visits), with no consistent gradient, and the ESI 1 group is small (26 visits). In other words, the choice of pathway is not strongly determined by acuity in this dataset, which is why acuity explains much less of the variation in length of stay than pathway does.

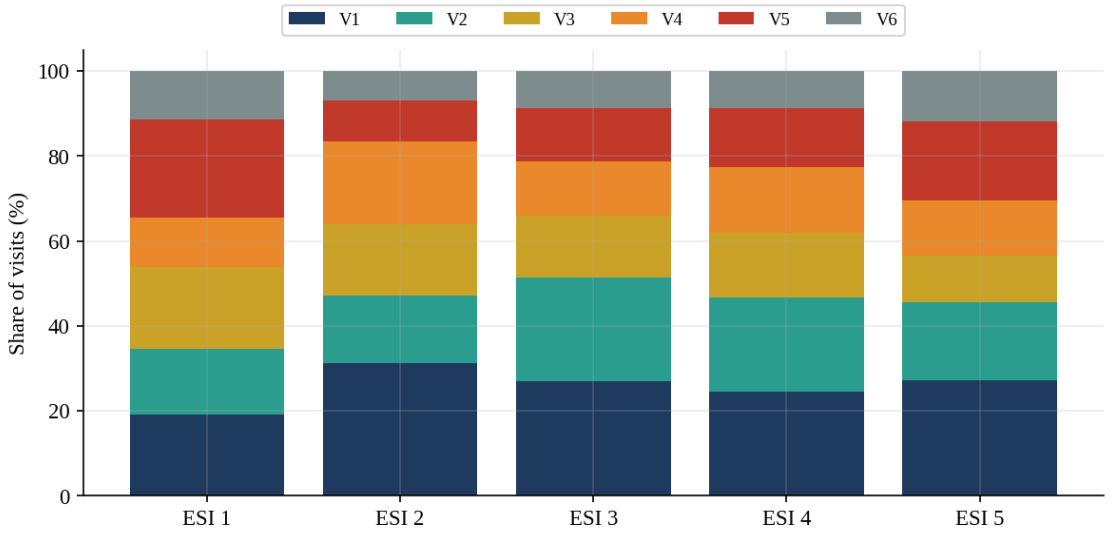


Figure 6.12: Share of the six pathway variants within each acuity level

Figure 6.13 decomposes the mean length of stay of each pathway into three stages: registration to triage, triage to first doctor contact, and doctor contact to discharge. The first two stages are nearly identical for all pathways (about 12.6 and 30.7 minutes on average); all of the difference between pathways arises after the doctor has seen the patient. This is an important observation

for improvement work: reducing the door-to-door time benefits every patient equally, whereas reducing the post-assessment stage requires pathway-specific measures.

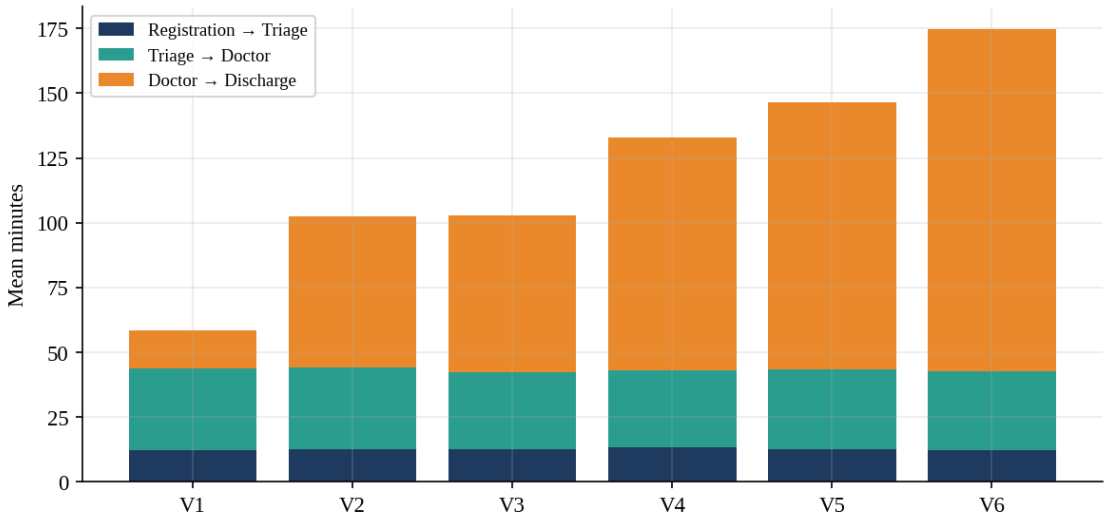


Figure 6.13: Mean time in each stage of the visit, by pathway variant

6.15 Critical versus Standard Cohorts

The Day 12 notebook split the visits into a critical cohort (ESI 1 and 2) and a standard cohort (ESI 3 to 5). Table 6.7 extends that comparison with the full set of timing and pathway measures. The two cohorts are of the same average age (55.0 years), so the comparison is not confounded by age.

Table 6.7: Comparison of the critical and standard cohorts

Measure	Critical (ESI 1–2)	Standard (ESI 3–5)
Visits	170	830
Mean age (years)	55.0	55.0
Registration to triage (min)	4.2	14.4
Door to doctor (min)	15.7	49.0
Doctor assessment to discharge (min)	62.4	64.5
Mean length of stay (min)	78.2	113.5
Median length of stay (min)	77	110
90th percentile length of stay (min)	134	168
Visits with a loopback (%)	25.9	23.0
Visits crossing midnight (%)	7.1	8.3
Visits with an X-ray (%)	53.5	59.4
Visits with laboratory work (%)	36.5	37.2

The table shows that the whole difference in length of stay arises before the doctor sees the patient. Critical patients wait 4.2 minutes for triage and 15.7 minutes in total for a doctor, against 14.4 and 49.0 minutes for the standard cohort, a difference of about 33 minutes. After the assessment both cohorts take about the same time (62.4 and 64.5 minutes), and the

difference in mean length of stay (78.2 versus 113.5 minutes, 35 minutes) is almost exactly the difference in door-to-doctor time.

Two further observations deserve mention. First, the critical cohort does not receive noticeably fewer or more diagnostics than the standard cohort (53.5% versus 59.4% with an X-ray; 36.5% versus 37.2% with laboratory work), which suggests that in this dataset acuity governs queue priority but not the clinical pathway. Second, the loopback rate of the critical cohort (25.9%) is slightly higher than that of the standard cohort (23.0%), but with 170 visits the difference is well within sampling variation. If the same pattern appeared in real data it would be worth examining, because rework in critically ill patients carries a higher clinical risk.

6.16 Summary of Exploratory Insights

Table 6.8: Insights from the exploratory analysis

Theme	Insight	Implication
Pathway	Pathway explains most of the variation in length of stay	Improvement efforts should target pathway steps, not patient groups
Acuity	Critical patients are seen in about 15 minutes; others in about 50	Prioritisation works; non-critical wait deserves attention
Rework	Rework is independent of age, acuity, weekday and shift	Fix at process level (paperwork), not through patient selection
Time of day	Arrivals are spread through the day; evening and night admissions cross midnight	Plan evening and night capacity
Physicians	Loopback rates differ ($p = 0.025$); length of stay does not ($p = 0.31$)	Review documentation practice; avoid individual judgement
Data	Few outliers; no genuine missing values	Typical of synthetic data; validate on real data

7. Process Mining and Workflow Analysis

This chapter presents the process-mining analyses developed in the prachi branch. Each analysis is a script that reads the event log, validates the required columns, sorts events by case and activity order, and writes a summary file and a short document.

7.1 Event-Log Validation

Before the full dataset was analysed, validate_event_log.py was tested on a 16-event, 3-case sample log. It checks that the required columns (case_id, activity, timestamp) exist, that none of them has missing values, that all timestamps are valid, that events can be ordered chronologically within each case, and that there are no exact duplicate rows. The sample passed all checks. The same validation logic was then applied at the start of every analysis script on the full 6,964-event log.

7.2 Activity Frequency and Distribution

Triage is the most frequent activity (1,235 events), because 235 patients were triaged twice. Registration, Doctor Assessment and Discharge each occur 1,000 times, once per visit. Diagnostic activities occur in subsets of visits: X-Ray, Radiology Result and Documentation Review in 584 visits (58.4%), Lab Collection and Lab Result in 371 visits (37.1%), and Doctor Reassessment in 235 visits (23.5%). Table 4.2 and Figure 4.1 show the full distribution.

7.3 Discovered Workflow (Process Map)

The directly-follows graph shows that every case starts with Registration (1,000 cases) and ends with Discharge (1,000 cases). All visits pass Registration, Triage and Doctor Assessment. After the assessment, 267 visits go directly to discharge, 584 visits go to X-Ray, and 149 visits go to the laboratory. A further 133 visits reach the laboratory after documentation review of imaging results. The re-triage loop (235 visits) leads from Radiology Result back to Triage and then to Documentation Review and Doctor Reassessment.

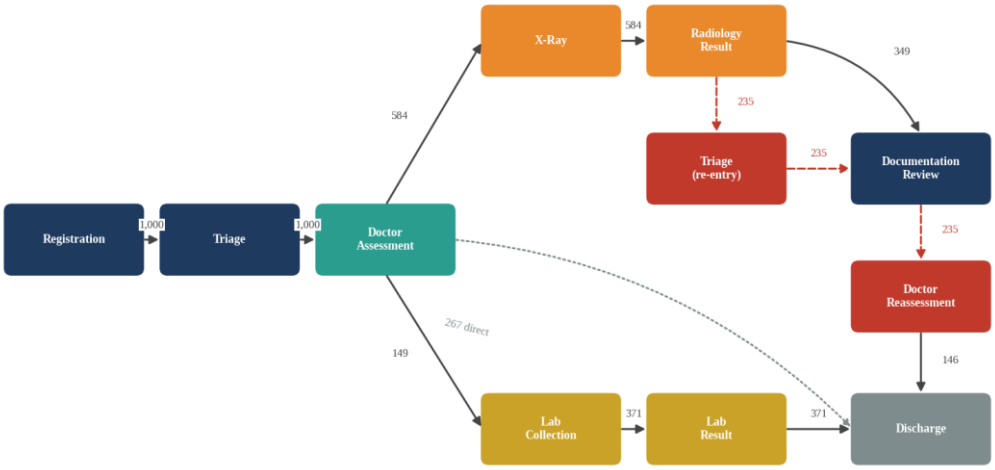


Figure 7.1: Process map of the CareFlow patient workflow with case counts. Dashed red arrows mark the loopback path. Not drawn: Documentation Review to Discharge (216), Documentation Review to Lab Collection (133), Doctor Reassessment to Lab Collection (89).

7.4 Transition Analysis and Waiting Time

Prachi's scripts identified 5,964 consecutive transitions of 16 unique types and computed for each the count and the average, median and maximum elapsed time (Table 7.1). Three transitions stand out. **Lab Collection to Lab Result** has the highest average elapsed time (32.56 minutes over 371 transitions). **Triage to Doctor Assessment** averages 30.74 minutes and occurs in all 1,000 visits, so it affects the largest number of patients. **X-Ray to Radiology Result** averages 22.42 minutes over 584 transitions.

Table 7.1: Activity transitions ranked by average elapsed time

From	To	Count	Mean (min)	Median (min)	Max (min)
Lab Collection	Lab Result	371	32.56	32.0	50
Triage	Doctor Assessment	1,000	30.74	29.0	60
X-Ray	Radiology Result	584	22.42	22.0	35
Documentation Review	Discharge	216	15.50	16.0	25
Doctor Reassessment	Discharge	146	15.37	16.0	25
Doctor Assessment	X-Ray	584	15.09	15.0	25
Lab Result	Discharge	371	14.87	14.0	25
Triage	Documentation Review	235	14.80	15.0	25
Doctor Assessment	Discharge	267	14.79	15.0	25
Registration	Triage	1,000	12.62	12.0	25
Doctor Assessment	Lab Collection	149	12.32	13.0	20
Documentation Review	Lab Collection	133	12.28	12.0	20
Doctor Reassessment	Lab Collection	89	12.15	13.0	20
Documentation Review	Doctor Reassessment	235	11.64	12.0	20
Radiology Result	Triage	235	8.67	8.0	15
Radiology Result	Documentation Review	349	6.56	7.0	11

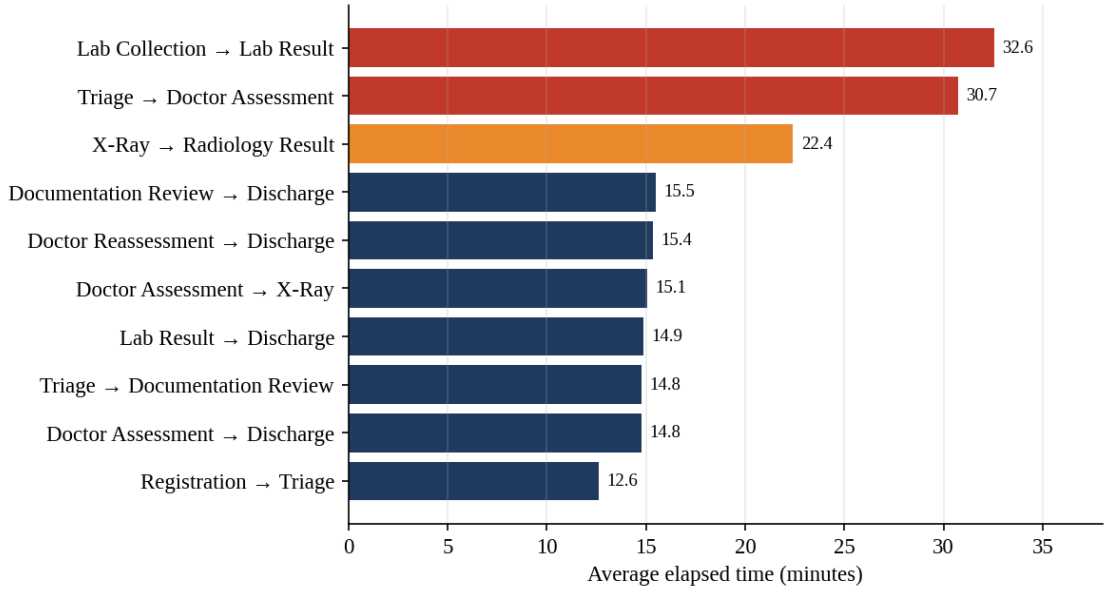


Figure 7.2: Ten transitions with the highest average elapsed time

Interpretation caution

Elapsed time between two recorded activities includes both waiting and the activity itself, so it does not by itself prove a bottleneck or its cause. The transitions above are the best candidates for further review; confirming them requires information such as queue lengths, staffing and equipment availability.

7.5 Pathway Variants

The 1,000 journeys follow exactly six complete pathways. The pathway comparison (Day 18) combines how common each pathway is with how long it takes. The direct fast track (V1) is the most common at 26.7% of visits and has the shortest mean duration, 58.6 minutes. Journey duration then rises with each additional diagnostic or rework step, to 174.6 minutes for the most complex pathway (V6), which includes radiology, laboratory and a reassessment loop.

Table 7.2: The six pathway variants

ID	Pathway	Cases	%	Mean (min)	Median	Min–Max
V1	Registration → Triage → Doctor Assessment → Discharge	267	26.7	58.6	58	12–107
V2	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Documentation Review → Discharge	216	21.6	102.3	102	52–158
V3	Registration → Triage → Doctor Assessment → Lab Collection → Lab Result → Discharge	149	14.9	102.9	102	44–157
V4	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Triage → Documentation Review → Doctor Reassessment → Discharge	146	14.6	132.8	134	73–192
V5	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Documentation Review → Lab Collection → Lab Result → Discharge	133	13.3	146.6	147	84–207
V6	Registration → Triage → Doctor Assessment → X-Ray → Radiology Result → Triage → Documentation Review → Doctor Reassessment → Lab Collection → Lab Result → Discharge	89	8.9	174.6	175	120–234

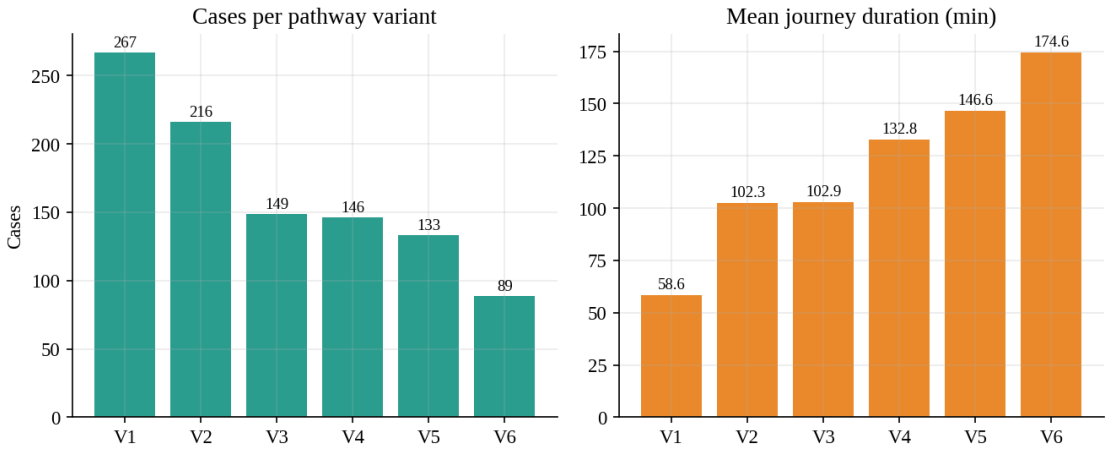


Figure 7.3: Cases and mean duration for each pathway variant

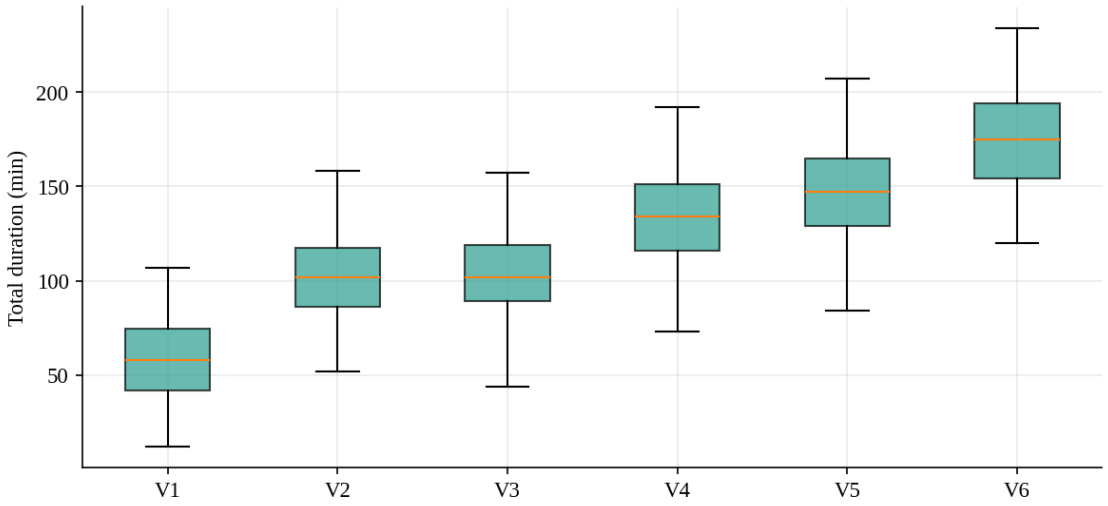


Figure 7.4: Spread of total duration within each pathway variant

The box plot shows that the ranges of neighbouring variants overlap, but the medians separate clearly: 58 minutes (V1), 102 minutes (V2 and V3), 134 minutes (V4), 147 minutes (V5) and 175 minutes (V6). Pathway therefore explains most of the variation in length of stay, which is consistent with the correlation analysis.

7.6 Loopback and Rework Analysis

Loopback analysis of the flagged events found 705 loopback-marked events in 235 visits (23.5%). Each of the three loop activities, re-triage, documentation review and doctor reassessment, contributes exactly 235 events, so every loopback case contains all three. The recorded loopback reason for all 235 cases is the same: *Missing intake paperwork*. Loopbacks occur only in variants V4 (146 cases) and V6 (89 cases), both of which include radiology; none occurs in the laboratory-only or direct pathways.

The additional time caused by a loopback can be estimated by comparing matched pathways that differ only by the loop. V4 (radiology with loop) takes 30.5 minutes longer than V2 (radiology without loop), a 29.8% increase. V6 takes 28.0 minutes longer than V5, both of which include radiology and laboratory, a 19.1% increase. Comparing all loopback visits (mean 148.6 minutes) with all non-loopback visits (94.9 minutes) gives a larger gap of 53.8 minutes, but this mixes the effect of the loop with the fact that loopback cases also include more diagnostics; the matched comparison is the fairer estimate.

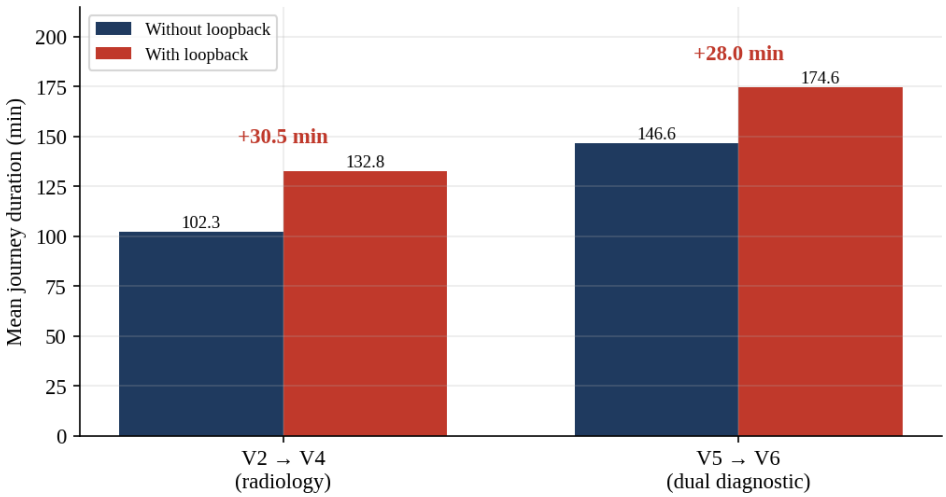


Figure 7.5: Extra journey time attributable to the loopback, from matched pathways

Applying the matched penalties to the loopback cases gives an indicative total of 30.48 × 146 + 28.01 × 89 ≈ 6,943 patient-minutes, or about 116 patient-hours, over the two-month period. This is a simple arithmetic estimate of time associated with rework in this dataset, not a forecast of savings.

7.7 Workflow Performance Summary and Recommendations

The scripts of Days 14 and 15 combine frequency and elapsed-time statistics into a ranked workflow summary and generate one review recommendation for each of the 16 transitions, together with suggested measures: median and 90th-percentile elapsed time for key transitions, number of affected cases, elapsed time by hour and acuity, and the frequency and documented reasons for repeated activities. The consolidated recommendations are presented in Chapter 9.

7.8 Activity Transition Matrix

Figure 7.6 presents the 5,964 observed transitions as a matrix: each row is the current activity and each column the next activity, and the cell shows how often that transition occurred. The matrix is sparse, which confirms that the process is highly structured: only 16 of the 100 possible pairs ever occur. Reading along the rows reveals the branching points. Doctor Assessment branches three ways (X-Ray, Lab Collection or Discharge), Radiology Result branches two ways (Documentation Review or re-triage), and Documentation Review branches three ways (Lab Collection, Doctor Reassessment or Discharge).

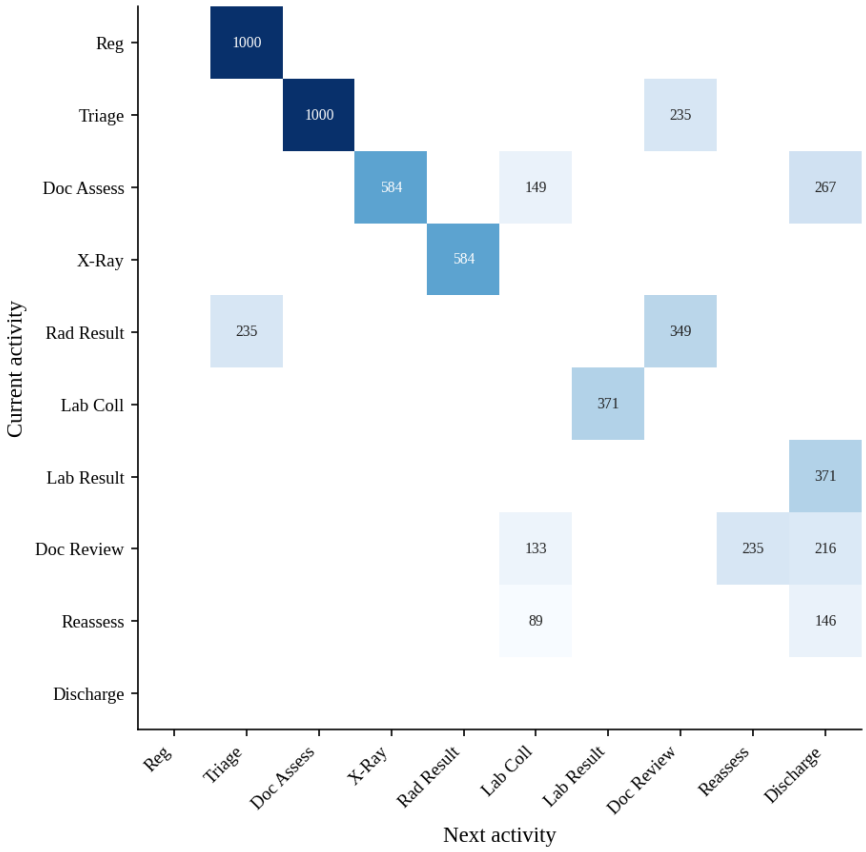


Figure 7.6: Transition counts between consecutive activities

Figure 7.7 shows the same matrix with the average elapsed time in each cell. The warmest cells are Lab Collection to Lab Result and Triage to Doctor Assessment, which confirms the ranking in Table 7.1; the coolest cells are Radiology Result to Documentation Review (6.6 minutes) and Radiology Result to Triage (8.7 minutes).

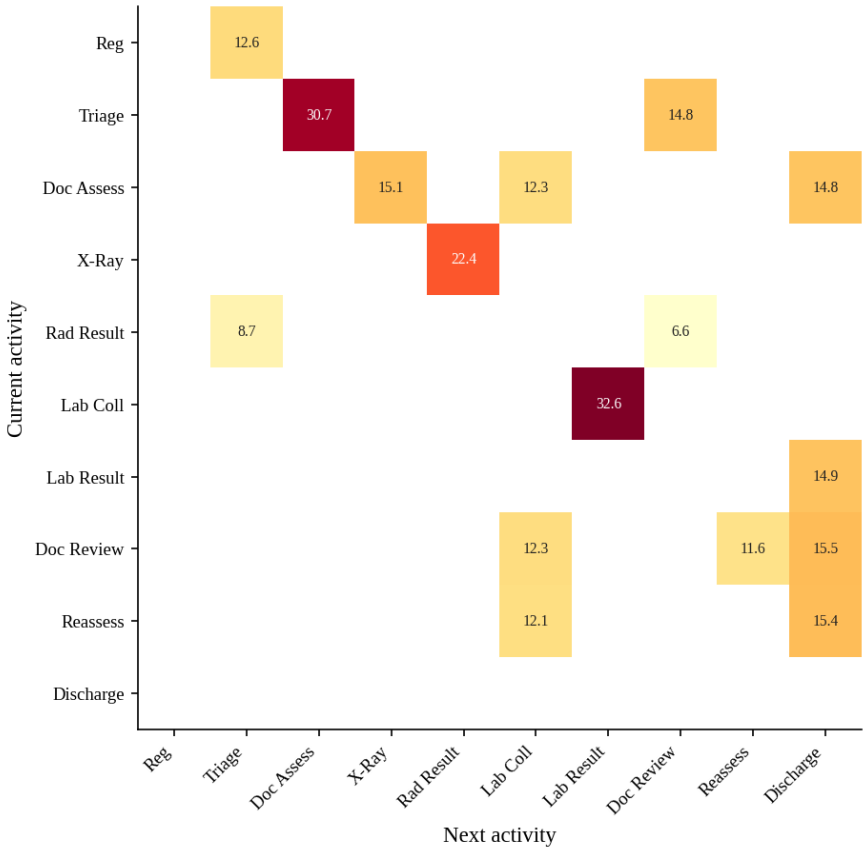


Figure 7.7: Average elapsed time (minutes) for each transition

7.9 Where the Time Goes

A transition with a long average time may still matter little if few patients experience it, and a short transition may matter a great deal if all patients experience it. Multiplying the count of each transition by its average elapsed time gives the total patient-minutes it consumes. Summed over all transitions the total is 107,496 patient-minutes, which equals the sum of the lengths of stay of the 1,000 visits and thus confirms the accounting. Figure 7.8 ranks the ten largest contributors.

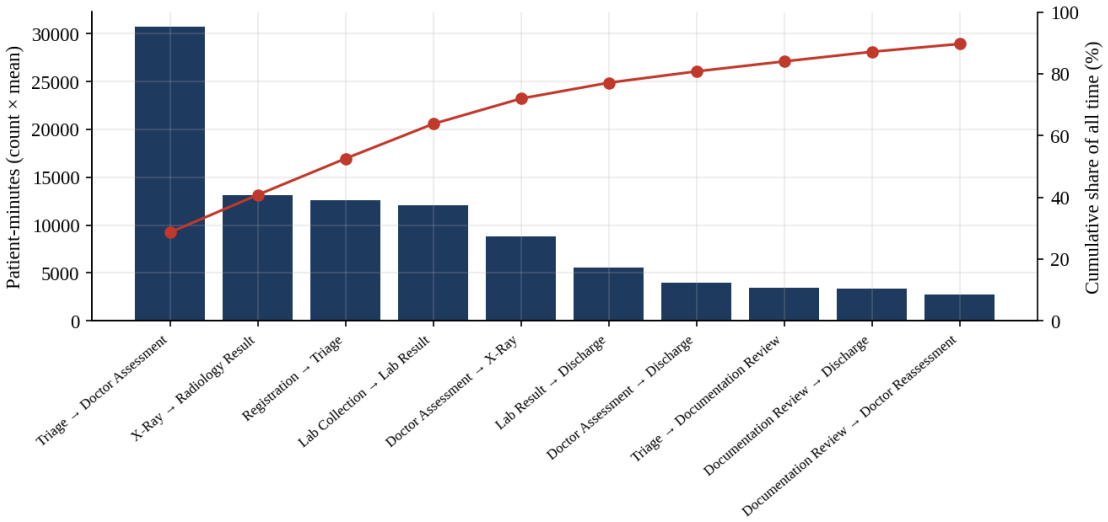


Figure 7.8: Patient-minutes consumed by each transition, with cumulative share

The result is striking. **Triage to Doctor Assessment** alone accounts for 30,738 patient-minutes, 28.6% of all time spent in the department. The next three transitions (X-Ray to Radiology Result, Registration to Triage, and Lab Collection to Lab Result) contribute 12.2%, 11.7% and 11.2% respectively, and the top four together account for 63.8% of the total. Laboratory turnaround has the longest average, but because only 37.1% of visits need laboratory work, it consumes less total time than the universal door-to-doctor interval. This ranking by total consumption, and not by average alone, should guide where improvement effort is spent first.

Table 7.3: Ten largest contributors to total patient time

Transition	Count	Mean (min)	Patient-minutes	Share (%)
Triage → Doctor Assessment	1,000	30.74	30,738	28.6
X-Ray → Radiology Result	584	22.42	13,095	12.2
Registration → Triage	1,000	12.62	12,623	11.7
Lab Collection → Lab Result	371	32.56	12,078	11.2
Doctor Assessment → X-Ray	584	15.09	8,810	8.2
Lab Result → Discharge	371	14.87	5,517	5.1
Doctor Assessment → Discharge	267	14.79	3,950	3.7
Triage → Documentation Review	235	14.80	3,478	3.2
Documentation Review → Discharge	216	15.50	3,349	3.1
Documentation Review → Doctor Reassessment	235	11.64	2,736	2.5

7.10 Profiles of the Six Pathways

Table 7.4 profiles the six pathways using the case table. The age and acuity of the patients differ little between pathways, which supports the observation that the pathway is largely independent of patient characteristics. The differences lie in the time taken after the doctor assessment and, consequently, in the total duration and the 90th percentile, which shows how long the slowest tenth of patients on each pathway stay.

Table 7.4: Profile of each pathway variant

ID	Cases	Mean age	Mean ESI	Critical (%)	Door to doctor	Doctor to discharge	Mean LOS	P90 LOS	Midnight
V1	267	58.0	3.27	18.7	43.8	14.8	58.6	89	10
V2	216	53.7	3.33	12.5	44.0	58.3	102.3	132	16
V3	149	53.7	3.22	19.5	42.2	60.7	102.9	137	12
V4	146	52.7	3.26	21.2	43.1	89.7	132.8	162	12
V5	133	53.4	3.38	15.0	43.5	103.1	146.6	181	18
V6	89	57.4	3.37	14.6	42.8	131.8	174.6	214	13

Table 7.5: Shift mix of each pathway variant (% of cases)

ID	Day	Evening	Night
V1	29.6	38.2	32.2
V2	28.2	36.6	35.2
V3	31.5	38.3	30.2
V4	32.9	32.2	34.9
V5	33.8	32.3	33.8
V6	36.0	33.7	30.3

7.10.1 V1: direct fast track

V1 contains 267 visits (26.7%), in which the patient is registered, triaged, assessed and discharged without diagnostics. It averages 4 events and 58.6 minutes in total, of which 14.8 minutes are spent between the assessment and discharge. Its 90th percentile is only 89 minutes. It is the benchmark against which the other pathways can be compared: any time beyond about an hour is attributable to diagnostics or rework.

7.10.2 V2: radiology without rework

V2 (216 visits) adds an X-ray, its report and a documentation review. The mean X-ray turnaround for these visits is 21.5 minutes, and the mean length of stay is 102.3 minutes, about 44 minutes more than V1. This is the pathway whose cost is the reference for estimating the effect of the loopback.

7.10.3 V3: laboratory only

V3 (149 visits) sends the patient to the laboratory directly after the doctor assessment. The mean laboratory turnaround is 33.8 minutes and the mean length of stay is 102.9 minutes, almost identical to V2, so for these two single-diagnostic pathways the choice of test does not change the total time.

7.10.4 V4: radiology with a rework loop

V4 (146 visits) follows the V2 route, but after the radiology result the patient is triaged again and passes through documentation review and a doctor reassessment before discharge. The

mean time from the first doctor contact to discharge rises to 89.7 minutes and the mean length of stay to 132.8 minutes, about 30 minutes more than V2. The mean loopback delay is 35.4 minutes, and 12 of these visits cross midnight.

7.10.5 V5: laboratory and radiology without rework

V5 (133 visits) combines imaging and laboratory work: the patient is X-rayed, the result is reviewed, and a laboratory test follows. Both diagnostic steps are performed in sequence, so the mean length of stay of 146.6 minutes is almost exactly the V1 baseline plus the extra time of the radiology route (V2) and the extra time of the laboratory route (V3). The 90th percentile is 181 minutes.

7.10.6 V6: laboratory and radiology with a rework loop

V6 (89 visits, 8.9%) is the most complex and the slowest pathway. It includes imaging, the loopback and the laboratory. The mean length of stay is 174.6 minutes, the 90th percentile 214 minutes, and 13 of the 89 visits cross midnight, a higher proportion than in any other pathway. Although V6 is the smallest group, it is the one in which a patient experiences the longest stay and the largest burden of repeated steps, and it is therefore a priority for the paperwork check recommended in Chapter 9.

7.11 Individual Patient Journeys

Aggregate statistics can hide what a single visit looks like. Figure 7.9 shows one example visit from each pathway as a timeline, with each block representing the time between one event and the next, coloured by activity category. The V1 patient spends most of the visit in administrative and clinical steps; the V2 and V3 patients spend most of their time in diagnostics; and the V4 and V6 patients show a block of red segments, which correspond to the loopback events, in the middle of the journey. The V5 patient, who needs both diagnostics, has the longest administrative phase at the start, which is a reminder that individual journeys vary widely around the pathway averages.

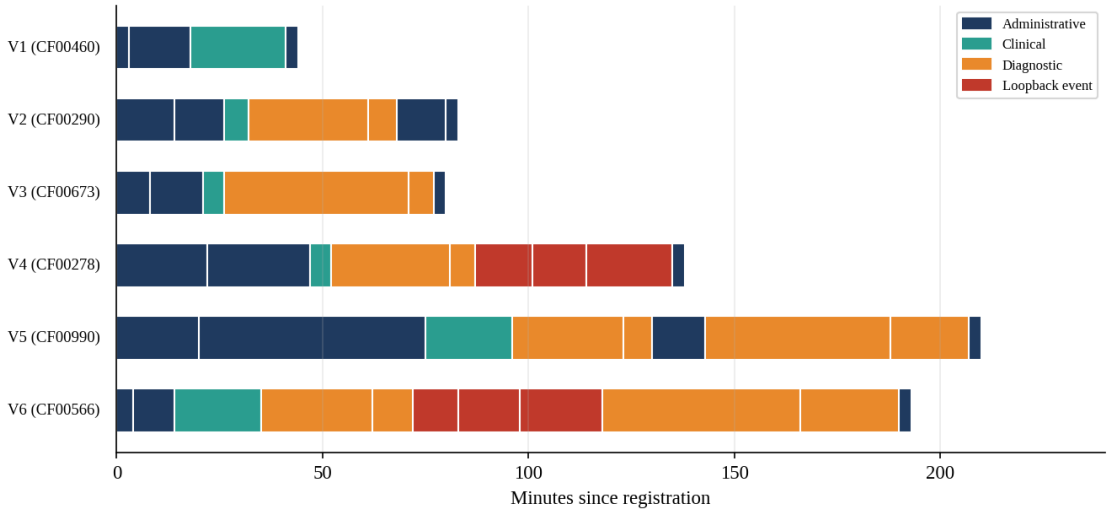


Figure 7.9: Example timelines of one visit from each pathway variant

7.12 Interpreting the Process Model

Several properties of the discovered model are worth noting. The model is **structured**: every visit starts at Registration and ends at Discharge, and the only cycle is the re-triage loop. It is **sequential**: although a real patient might have laboratory and radiology tests in parallel, the log records them as a sequence, so the DFG cannot reveal concurrency. It is **homogeneous**: the pathway distribution is similar across acuity levels, age groups and shifts, which is unusual in real emergency departments, where critical patients typically follow different routes from minor cases. These properties are consistent with synthetic data generated from a small set of templates and should be borne in mind when the same method is applied to real logs, which will almost certainly show more variants, more concurrency and more noise.

8. Dashboards and Visualisation

8.1 Power BI Report

The sagar branch contains four Power BI files, each holding one 1280 × 720 report page built from careflow_cases_imputed.csv and careflow_events_scaled.csv. All pages share a common layout with a dark header, slicers for acuity, age group, shift and weekend, and a footer stating that the data are synthetic and cover 1 January to 2 March 2026.

Table 8.1: Power BI report pages

Page (file)	Key visuals
Overview (CareFlow_Emergency_Dashboard)	Cards for Total Cases, Average Duration, Average Door to Doctor and Loopback %; weekly line chart of cases; column chart of average duration by shift; bar charts of average duration by age group and of the top five variants
Process and bottlenecks (CareFlow_Process_and_Bottlenecks)	Bar chart of average wait before each activity; bar chart of the top five variants
Loopbacks and outliers (CareFlow_Loopbacks_and_Outliers)	Cards for Loopback Cases, Loopback % and Outlier Cases; card for the top loopback reason; bar chart of loopback % by acuity; scatter chart of age versus duration coloured by acuity; page navigator
Doctors and workloads (CareFlow_Doctors_and_Workloads)	Bar chart of average duration by doctor; matrix of average duration by doctor and shift; slicers for acuity, shift, weekend and age group

The Day 18 notes record the integration of the Loopbacks and Outliers page into the pipeline and describe its purpose as isolating pathways with abnormally long transit and waiting times and mapping recurring returns between triage and diagnostic steps.

8.2 Streamlit Interactive Dashboard

The Day 17 Streamlit application (careflow_interactive_dashboard.py) reads the processed summary CSV files and presents key workflow metrics as KPI tiles, charts of activity frequency, charts of the six patient pathways, a ranking of transitions with the highest average elapsed time, and summary tables. Because it reads precomputed summaries, it requires the processed files to be present and does not recompute the analysis.

8.3 Static Visual Outputs

The Day 16 visualisation script produced three charts from the existing summaries: an activity frequency chart, a patient pathway chart and a chart of the ten transitions with the highest average elapsed time. A pathway-details file maps each pathway label to its full activity sequence. The charts in this report were regenerated from the same data for consistent styling.

8.4 Metric Definitions

The Power BI pages present a small set of measures whose definitions are given in Table 8.2. Each can be reproduced directly from the case table, which is how the figures in this report were verified.

Table 8.2: Measures shown on the dashboard pages

Measure	Definition	Value in the full data
Total Cases	Count of distinct case_id	1,000
Average Duration	Mean of total_duration_min	107.5 min
Average Door to Doctor	Mean of door_to_doctor_min	43.4 min
Loopback Cases	Count of cases with loopback_flag = 1	235
Loopback %	Loopback Cases divided by Total Cases	23.5%
Outlier Cases	Count of cases with duration_outlier_flag = 1	1
Average wait before activity	Mean of minutes_since_prev grouped by activity	Table 4.4

8.5 How to Read the Report Pages

- **Overview page.** Start with the four cards: they give the volume, the typical length of stay, the typical wait for a doctor and the share of visits with rework. The weekly line shows whether volume is stable; the shift and age bars show whether any group stays longer; the variant bars show which pathways dominate.
- **Process and bottlenecks page.** The bar chart of wait before each activity shows where time accumulates between steps. Use the slicers to see whether the ranking changes for high-acuity patients or for a particular shift.
- **Loopbacks and outliers page.** The cards show the extent of rework and the number of flagged outliers; the bar chart of loopback rate by acuity shows whether any acuity group is affected more; the scatter plot shows whether any patients are extreme in both age and duration.
- **Doctors and workloads page.** The bar chart ranks physicians by mean duration, and the matrix breaks this down by shift. Use it to prompt a review of workload balance, not to evaluate individuals (Section 6.6).

8.6 Running the Streamlit Dashboard

The interactive application is started from the repository root once the processed summary files exist. It uses Streamlit, which serves the dashboard locally in a web browser.

Starting the interactive dashboard

```
pip install streamlit pandas plotly
streamlit run src/careflow_interactive_dashboard.py
```

8.7 Design Principles

Each page answers one group of questions, so that a manager can find the relevant view quickly. The same acuity, age group, shift and weekend slicers appear on every page with the same meaning. The footer of every page states that the data are synthetic and gives the period covered. Warm colours (red, orange) mark rework and delay, and cool colours (blue, teal) mark normal flow, as in the figures of this report.

9. Results, Insights and Recommendations

9.1 Summary of Key Findings

Table 9.1: Principal findings

#	Finding	Evidence
1	Pathway complexity is the main driver of length of stay	Six pathways with median duration from 58 to 175 minutes; correlation of LOS with loopback r = 0.524
2	Laboratory turnaround is the slowest transition	Lab Collection to Lab Result averages 32.6 minutes (median 32), 45% longer than X-Ray to Radiology Result (22.4)
3	Door-to-doctor wait is the largest pre-diagnostic delay and is acuity-dependent	Triage to Doctor Assessment averages 30.7 minutes; door-to-doctor 14 to 16 minutes for ESI 1 and 2 versus 49 to 50 for ESI 3 to 5
4	Nearly one visit in four involves rework, all with the same recorded reason	235 visits (23.5%); reason “Missing intake paperwork” for all; only in radiology pathways V4 and V6
5	Rework adds roughly 28 to 30 minutes to a journey	Matched comparisons V4 versus V2 (+30.5) and V6 versus V5 (+28.0)
6	Overnight boarding arises only from evening and night admissions	81 visits cross midnight: 36 evening, 45 night, 0 day
7	Loopback rates differ between physicians	12.3% (DR11) to 34.1% (DR12); chi-square p = 0.025; LOS differences not significant
8	A leakage-free modelling dataset is ready	41 arrival-time features, four targets, 16 of 16 validation checks passed

9.2 Recommendations

The recommendations below link each finding to an action and to a measure by which the action can be evaluated. They are derived from the analysis of this dataset and should be validated with operational information before implementation.

1. **Reduce the door-to-doctor wait for non-critical patients.** The 30.7-minute triage-to-assessment interval and the 49 to 50 minute wait for ESI 3 to 5 patients suggest a fast-track route for low-complexity cases and dynamic queue routing. *Measure:* median and 90th-percentile triage-to-assessment time by hour and acuity.
2. **Review laboratory turnaround.** Examine sample transport, analyser queues and result-recording practice, and consider point-of-care testing for high-acuity patients. *Measure:* median and 90th-percentile Lab Collection to Lab Result time.
3. **Eliminate the documented cause of rework.** Since all 235 loopbacks are recorded as missing intake paperwork, introduce a completeness check at registration or triage so that required paperwork is confirmed before the patient proceeds to imaging. *Measure:* loopback rate and the share of cases with a repeated triage.
4. **Shorten radiology reporting.** The 22.4-minute reporting interval and the fact that every loop begins after a radiology result make the imaging-to-review handover a priority for review of queues and report timestamps.

5. **Plan evening and night capacity.** Because every midnight crossing occurs in evening or night admissions, use the cyclical arrival features to forecast late-day load and to reserve beds and staff before shift handover.
6. **Review physician-level variation as a process question.** Examine whether differences in loopback rate reflect case mix, documentation habits or handover practice, using a larger sample before drawing conclusions.
7. **Deploy predictive scoring and monitoring.** Use the model-ready feature store to build length-of-stay and loopback predictors, and keep the Power BI and Streamlit dashboards as a continuous monitoring view for clinical managers.

9.3 Indicative Impact

The rework associated with the 235 loopback visits corresponds to about 116 patient-hours over the study period (Section 7.6). Reducing the 30.7-minute triage-to-doctor interval or the 32.6-minute laboratory interval would act on a larger number of patients: the first interval is present in all 1,000 visits and the second in 371 visits. Quantifying the benefit of any specific intervention would require operational data that are not part of this dataset.

9.4 Findings by Stakeholder

Different readers need different parts of the analysis. Table 9.2 organises the findings by the audience most likely to act on them.

Table 9.2: Key messages for each stakeholder group

Stakeholder	What they need to know	Key figures
ED clinical manager	Where time is lost and what rework costs	Triage to doctor 30.7 min (28.6% of all patient time); rework adds 28 to 30 min in 23.5% of visits
Triage and registration nurses	Why patients return to triage	All 235 loopbacks are recorded as missing intake paperwork
Laboratory manager	Whether turnaround is a problem	Lab Collection to Lab Result averages 32.6 min, the longest interval; 90th percentile 48 min
Radiology manager	Reporting time for imaging	X-Ray to Radiology Result averages 22.4 min over 584 visits
Bed and staffing planners	When pressure on overnight capacity arises	81 visits cross midnight, all admitted in the evening or at night
Data and IT team	What data would improve the analysis	Queue and start-of-service timestamps; multi-department events; real data

9.5 Illustrative What-If Scenarios

The scenarios in Table 9.3 show the arithmetic of what would happen to total patient time if the average elapsed time of a transition were reduced by 10%, 20% or 30%, holding everything else constant. The numbers are illustrations of scale and not predictions: they assume that the

reduction applies to every patient who passes through the transition, and they ignore secondary effects such as shorter queues for other steps.

Table 9.3: Patient-hours saved over the two-month period for hypothetical reductions

Transition	Visits	Mean (min)	10% reduction	20% reduction	30% reduction
Triage → Doctor Assessment	1000	30.7	51 h	102 h	154 h
Lab Collection → Lab Result	371	32.6	20 h	40 h	60 h
X-Ray → Radiology Result	584	22.4	22 h	44 h	65 h

For comparison, eliminating the rework entirely would, by the matched-pathway estimate in Section 7.6, release about 116 patient-hours. A 20% reduction in the triage-to-doctor interval would release a comparable amount (about 102 patient-hours) and would benefit all 1,000 visits rather than 235, which illustrates why both interventions deserve attention.

9.6 Implementation Roadmap

Figure 9.1 sketches an indicative sequence for the recommendations. Quick wins that need no new data, such as the paperwork checklist, are placed first. Measures that require additional data collection, such as queue and start-of-service timestamps, run in parallel because they enable the later process redesign and model work. The durations are indicative and would need to be adapted to the resources of a real department.

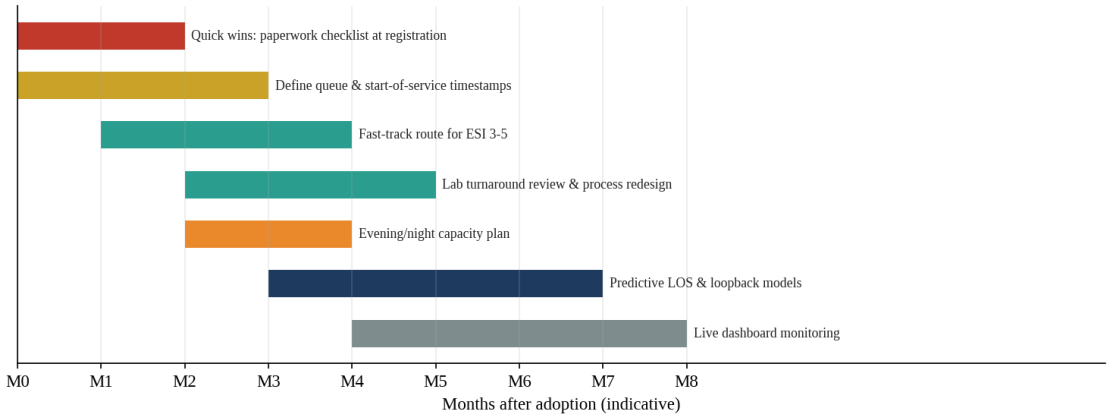


Figure 9.1: Indicative roadmap for implementing the recommendations

9.7 Monitoring Framework

Table 9.4: Key indicators for monitoring improvement

Indicator	Definition	Baseline in this dataset	Desired direction
Door-to-doctor time (median, P90)	Registration to first doctor assessment	Mean 43.4 min; triage to doctor median 29, P90 55	Decrease
Door-to-doctor for ESI 1 and 2	Same, for critical patients	About 14 to 16 min	Maintain
Loopback rate	Share of visits with a repeated triage	23.5%	Decrease
Laboratory turnaround (median, P90)	Lab Collection to Lab Result	Median 32, P90 48 min	Decrease
Radiology reporting time	X-Ray to Radiology Result	Median 22, P90 33 min	Decrease
Length of stay (median, P90)	Registration to discharge	Median 105 min; mean 107.5 min	Decrease
Midnight crossings	Share of visits ending after midnight	8.1%	Decrease

9.8 Dependencies and Risks to Implementation

- **Data availability:** several recommendations depend on timestamps that the current log does not contain; without them, waiting cannot be separated from service time.
- **Clinical safety:** any change to triage or routing must be reviewed by clinicians so that the priority of critical patients is preserved.
- **Change management:** a paperwork checklist is easy to specify but depends on adoption by registration staff; the loopback rate is the indicator of whether it works.
- **Generalisability:** because the findings are based on synthetic data, they must be confirmed on real data before any operational change.

10. Testing, Validation and Reconciliation

10.1 Pipeline Validation

The preprocessing pipeline is self-testing: its 16 checks, listed in Table 5.5, cover the chronological split, structural missingness, completeness after imputation, the statistical properties of the scaled data, reversibility of the scaling, row-count integrity of the event table and absence of nulls in the feature store. All checks passed.

10.2 Cross-Checking Results Across Team Members

Because the three analytical branches work from the same event log, their outputs can be compared. The key quantities agree exactly across the branches (Table 10.1), which supports the integrity of the shared data and of the separate analysis code.

Table 10.1: Quantities independently reproduced in more than one branch

Quantity	Value	Reproduced in
Cases and events	1,000 and 6,964	All branches
Distinct activities	10	Preprocessing, EDA, process mining
Pathway variants and the most common share	6; 26.7% (267 cases)	EDA notebook (Day 7), process-mining scripts (Days 7, 11, 18)
Loopback cases and events	235 and 705	EDA notebook (Day 7), process mining (Day 4), preprocessing
Lab Collection to Lab Result mean	32.56 min	Process mining (Days 8, 13, 14), preprocessing (Lab Result wait)
Triage to Doctor Assessment mean	30.74 min	Process mining (Days 8, 13, 14), preprocessing (Doctor Assessment wait)
Mean length of stay	107.5 min	EDA notebooks (Days 6, 16), case table

10.3 Reconciliation Notes

While consolidating the branches for this report, the shared figures were recomputed from the processed files. Several statements in the intermediate documents did not match the data exactly. The corrected values used in this report are summarised below so that the team can align the source documents.

Table 10.2: Differences found and the values used in this report

Item	As stated in a source document	Value from the data
Average “wait time” in the Day 10 audit	64.1 min	This is the mean assessment-to-discharge time; mean total length of stay is 107.5 min
Hourly “patient count” in the Day 13 notebook	About 250 to 350 per hour	These are event counts (sum 6,964); patient arrivals per hour range from 28 to 60
Outliers in the Day 16 summary	80 cases (8.0%)	One case above the IQR fence for total duration; none for assessment-to-discharge
Bottleneck labels in the Day 16 summary	Triage to Lab Result; Registration to Doctor Assessment	Recorded transitions are Lab Collection to Lab Result (32.56) and Triage to Doctor Assessment (30.74); Registration to Doctor Assessment is the sum of two transitions (43.4 min)
Midnight crossings	37 evening, 44 night	36 evening, 45 night (81 in total)
Delay caused by rework	More than 45 minutes	28.0 to 30.5 minutes in matched pathways (53.8 minutes in the unmatched comparison)
ICU handoffs in the stakeholder summary	Bottlenecks in Emergency to ICU handoffs	The dataset contains one department and no ICU events; this statement cannot be supported by the supplied data
Imputation advantage	39.2% better than “baseline imputers”	39.2% lower NRMSE than mean imputation; 15.6% lower than KNN

10.4 Test-Case Matrix

Table 10.3 lists the principal tests applied to the pipeline and the analyses, the condition that was expected, and the outcome. The first twelve correspond to the automated checks and the remainder to manual verification during the preparation of this report.

Table 10.3: Test cases and outcomes

ID	Test	Expected	Result
T1	Chronological split has no overlap	Latest training start not after earliest test start	Pass
T2	Structural gaps coincide with indicator flags	Missing exactly when the indicator is zero	Pass
T3	No missing values remain after imputation	Zero nulls in all case columns	Pass
T4	Standardised training mean and standard deviation	Mean 0 (within 1.3×10^{-16}) and SD 1	Pass
T5	Normalised training range	Minimum 0 and maximum 1	Pass
T6	Inverse transformation	Original values recovered	Pass
T7	Event table row count	6,964 rows retained after scaling	Pass
T8	No nulls in the feature store	Zero nulls in 41 columns	Pass
T9	Event log schema validation	Required columns present; timestamps valid	Pass on sample and full log
T10	Every case starts with Registration and ends with Discharge	1,000 of 1,000	Pass
T11	Sum of transition times equals sum of lengths of stay	107,496 minutes on both sides	Pass
T12	Loopback events equal three times loopback cases	$705 = 3 \times 235$	Pass
T13	Matched variant counts across branches	6 variants; 267, 216, 149, 146, 133, 89	Pass
T14	Reproduce reported averages from processed files	Agreement with notebooks	Mostly pass; discrepancies in Table 10.2

10.5 Reproducibility Checklist

- Install Python 3.10 or later and the packages listed in Section 3.9.
- Place the raw event log and case attributes in the data folder as described in the README.
- Run the preprocessing script and confirm that the console reports 16 passed checks.
- Open the notebooks in order from Day 3 to Day 16; each reads the processed files and writes its own summary.
- Run the process-mining scripts of the prachi branch and confirm that the summary files match Table 7.1 and Table 7.2.
- Open the Power BI files and refresh the data source if the file paths differ on the local machine.

11. Ethics, Privacy and Governance

Healthcare analytics deals with information about people at vulnerable moments. Even when the data are synthetic, as here, the practices adopted in a project should be those that would be needed with real data. This chapter summarises the privacy, fairness and governance considerations that applied to CareFlow and that should be carried forward.

11.1 Privacy and Data Protection

The project followed the data-governance rule in the assignment: only public, de-identified or synthetic data are used, and confidential patient information is never published. In the supplied data, patients are identified only by case codes (for example CF00278) and physicians by codes (DR01 to DR12); there are no names, dates of birth, addresses or other direct identifiers. Ages are recorded as whole years. Even so, a combination of age, arrival time and acuity could in principle single out a patient in a small department, so with real data additional safeguards would be required: aggregation of small groups, coarser time resolution in shared reports, access control to the case-level tables, and a formal data-protection assessment.

11.2 Fairness and Individual Accountability

Two analyses in this report could, if misused, disadvantage individuals or groups. The first is the comparison of physicians. The loopback rates differ between physicians (Section 6.6), but the samples are small, the case mix may differ, and the cause is recorded as missing intake paperwork, which is not obviously under the physician's control. The comparison is therefore presented as a prompt for process review and not as a performance ranking. The second is the acuity-based difference in waiting time. Prioritising critical patients is clinically appropriate, but the long wait of non-critical patients should be monitored so that it does not become unsafe or unacceptable; the monitoring indicators in Chapter 9 include both groups.

11.3 Responsible Use of Predictive Models

The feature store was designed for predicting length of stay, loopback and midnight crossing. If such models are built, several safeguards apply. Predictions should support, not replace, clinical judgement and must never be used to delay care. The models must be evaluated on chronologically later data and monitored for drift, since a change in the case mix or in the process would reduce their accuracy. Features that encode physician identity should be used cautiously, because they could turn a prediction into an implicit evaluation of individuals. Finally, the limitations of synthetic training data must be disclosed to anyone who reads a model's output.

11.4 Governance of Dashboards and Reports

- **Ownership:** each dashboard page and each derived dataset should have a named owner who is responsible for refreshing and correcting it.
- **Versioning:** the scaler parameters and the processed files are versioned together in the repository, so any figure can be traced to the data and code that produced it.
- **Change control:** changes to metric definitions (Table 8.2) should be reviewed and recorded so that figures remain comparable over time.
- **Transparency:** the reconciliation of discrepancies (Chapter 10) is part of the record and should be kept with the report.

11.5 Checklist Before Applying the Method to Real Data

Moving from the synthetic log to a real hospital event log changes the risk profile of the project. Before any real data are loaded, the following steps should be completed and documented.

1. **Authorisation:** obtain written approval from the hospital's data-protection officer or ethics committee, specifying the purpose, the data fields and the retention period.
2. **De-identification:** remove direct identifiers, replace case and staff identifiers with irreversible codes, and consider generalising ages and coarsening timestamps.
3. **Minimum necessary data:** use only the fields needed for the analysis; the five event fields plus acuity, age and physician code are sufficient for everything in this report.
4. **Secure environment:** keep case-level data on approved systems with access control and logging, and share only aggregated results.
5. **Small-number suppression:** do not publish groups with very few cases, such as a single physician-shift combination, in shared dashboards.
6. **Validation before use:** repeat the reconciliation of Chapter 10 on the real data and confirm that the 16 pipeline checks pass before any result is reported to clinical staff.

12. Limitations and Future Scope

12.1 Limitations

- **Synthetic data.** The event log is synthetic and covers one department for two months. Patterns such as the single recorded loopback reason, the absence of outliers and the sharp acuity split are cleaner than real hospital data and should not be generalised.
- **Elapsed time is not waiting time.** Transition times include the activity itself; true waiting time needs queue and start-of-service timestamps.
- **No genuine missing data.** The benchmark of imputation methods was run on artificially masked values because no genuine gaps existed.
- **Small physician samples.** Between 64 and 106 visits per physician limit the strength of comparisons.
- **No predictive model was trained.** The feature store and targets are ready, but model training and evaluation fall outside the work completed in this assignment; the only fitted model was the single-feature random forest of Day 14.
- **Single department.** The Day 10 audit records one unique department, so flows between departments and to inpatient wards could not be analysed.

12.2 Future Scope

- Train and evaluate length-of-stay, loopback and midnight-crossing models on the model-ready store, using chronological validation and the metrics suggested in Section 5.8.
- Add conformance checking against a reference pathway and variant clustering with PM4Py.
- Collect true queue and start-of-service timestamps to separate waiting from activity time.
- Extend the log to include real or realistically simulated multi-department data, including admission and ICU handoffs.
- Automate the pipeline, from raw log to dashboards, and add real-time process-mining monitoring for clinical managers.

13. Conclusion

CareFlow demonstrates how process mining and data-science preprocessing can turn an emergency department event log into operational insight. From 6,964 events across 1,000 visits, the team reconstructed six patient pathways, showed that pathway complexity explains most of the variation in length of stay, located the slowest transitions (laboratory turnaround at 32.6 minutes and triage-to-doctor at 30.7 minutes), and quantified that rework, present in 23.5% of visits and always recorded as missing intake paperwork, adds about 28 to 30 minutes to the journeys it affects.

In parallel, a leakage-free preprocessing pipeline delivered imputed, standardised and normalised datasets, a 41-feature arrival-time store with four targets, persisted scaler parameters and a 16-point validation suite, all passing. Interactive Power BI and Streamlit dashboards make the results accessible to non-technical users, and the recommendations connect each finding to an action and a measurable indicator.

The main caveat is that the data are synthetic; the value of the project lies in a method that can be applied directly to real, de-identified hospital event logs. The next step is to apply the same pipeline to such data and to train and evaluate the predictive models for which the feature store has been prepared.

13.1 Learning Outcomes

Beyond the technical results, the assignment developed skills that are central to professional data analytics.

- **Process mining in practice:** reconstructing traces from an event log, building and interpreting a directly-follows graph, and ranking transitions by both average time and total time consumed.
- **Leakage-aware data preparation:** designing a pipeline in which the split, imputation and scaling are ordered so that no information from the future is used, and verifying it with automated checks.
- **Critical verification:** recomputing figures from the data, discovering that several published summary numbers did not match, and correcting them transparently rather than repeating them.
- **Statistical reasoning:** choosing appropriate tests, recognising the limits of small samples and distinguishing statistical significance from practical importance.
- **Visual communication:** designing charts and dashboard pages that answer one question each and are readable by non-technical managers.
- **Collaboration:** working in parallel branches of a shared repository, agreeing on file formats and metric definitions, and cross-checking each other’s results.

13.2 Closing Remarks

The CareFlow project shows that much of the value of hospital data lies in the order and timing of events. The next step is to carry the method from synthetic data to real, de-identified data and to complete the predictive models for which the foundation has been laid.

References

[1] W. M. P. van der Aalst, Process Mining: Data Science in Action, 2nd ed. Berlin, Germany: Springer, 2016.

[2] A. Berti, S. J. van Zelst and W. M. P. van der Aalst, “Process Mining for Python (PM4Py): Bridging the Gap Between Process- and Data Science,” arXiv:1905.06169, 2019.

[3] N. Gilboy, P. Tanabe, D. Travers, A. M. Rosenau and others, Emergency Severity Index (ESI): A Triage Tool for Emergency Department Care, Version 4, Implementation Handbook. Rockville, MD: Agency for Healthcare Research and Quality, 2012.

[4] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[5] Infotact Solutions, Project Assignment Letter, Ref. ITS/DA/1176: CareFlow: Clinical Pathway Process Mining, 1 October 2026.

[6] CareFlow project repository, branches main, akshar, sagar and prachi (README files, notebooks, scripts and processed datasets).

Appendix A: Data Dictionary of the Case Table

Table A.1: Columns of careflow_cases_imputed.csv (1,000 rows, 32 columns)

Column	Description
case_id	Visit identifier
case_start, case_end	Registration time and discharge time
total_duration_min	Length of stay in minutes
n_events, n_unique_activities	Number of events and number of distinct activities in the visit
variant_id, variant	Pathway variant (V1 to V6) and its activity sequence
age, age_group	Patient age and band (18-39, 40-59, 60+)
acuity	ESI level, 1 to 5
resource, department	Attending physician (DR01 to DR12) and department
has_xray, has_lab, has_reassessment	Indicators that the procedure occurred
loopback_flag, loopback_reason	Rework indicator and recorded reason
registration_to_triage_min	Registration to first triage
door_to_doctor_min	Registration to first doctor assessment
xray_to_result_min, lab_turnaround_min	Imaging and laboratory turnaround; 0 when the procedure did not occur
assessment_to_discharge_min, loopback_delay_min	Assessment to discharge; delay attributable to reassessment (0 if none)
start_hour, start_weekday, start_month, is_weekend, shift	Arrival hour, weekday, month, weekend flag and shift
ends_next_day, duration_outlier_flag	Midnight-crossing flag and IQR outlier flag
split	Chronological train or test label

Appendix B: Daily Progress Log

The table below summarises the chronological record of the work as documented in the repository, merging the three workstreams.

Table B.1: Documented work by day

Day	Preprocessing and consolidation	Exploratory analysis	Process mining
1	Kickoff, environment		
2	Cleaning, validation		
3	Master dataset	Baseline EDA	Event-log validation
4	Acuity stratification	Demographics	Loopback analysis
5	Activity frequency	Activity frequency	Transition analysis
6	Duration analysis	LOS timeline	Workflow discovery
7	Trace mining	Trace mining	Pathway variation
8	Transition matrix	Transition matrix	Waiting-time analysis
9	Acuity correlation	Acuity correlation	Activity frequency
10	Mid-project audit	Mid-project audit	Activity distribution
11		Outlier detection	Journey sequence analysis
12		Cohort segmentation	Journey transitions
13		Peak loads	Bottleneck ranking
14		Feature significance	Performance summary
15		Manager insights view	Recommendations
16		Final metrics summary	Static charts
17			Streamlit dashboard
18	Dashboard integration notes	Dashboard notes	Journey comparison

Appendix C: Code Excerpts

C.1 Chronological split (careflow_impute_scale.py)

```
# ------------------------------------------------ 1. CHRONOLOGICAL SPLIT
cases = cases.sort_values("case_start", kind="stable").reset_index(drop=True)
cutoff = cases["case_start"].quantile(0.8)
cases["split"] = np.where(cases["case_start"] <= cutoff, "train", "test")
tr = cases["split"].eq("train")
split_map = cases.set_index("case_id")["split"]
log("split is chronological (max train start <= min test start)",
    cases.loc[tr, "case_start"].max() <= cases.loc[~tr, "case_start"].min(),
    f"train={tr.sum()} test={(~tr).sum()} cutoff={cutoff}")
```

C.2 Structural missingness handling (careflow_impute_scale.py)

```
STRUCT = {"xray_to_result_min": "has_xray", "lab_turnaround_min": "has_lab",
"loopback_delay_min": "has_reassessment"}
imp = cases.copy()
for col, flag in STRUCT.items():
    imp.loc[imp[flag] == 0, col] = np.nan          # make the structural gap explicit
    log(f"{col} is NaN exactly when {flag} == 0", (imp[col].isna() == (imp[flag] ==
0)).all())
    imp[col] = imp[col].fillna(0.0)
```

C.3 Elapsed time between consecutive activities (process_bottleneck_analysis.py)

```
# Sort patient events
df = df.sort_values(["case_id", "activity_order"]).copy()


# Identify next activity and its timestamp
df["next_activity"] = df.groupby("case_id")["activity_name"].shift(-1)
df["next_timestamp"] = df.groupby("case_id")["timestamp"].shift(-1)


# Calculate elapsed minutes between consecutive activities
df["elapsed_minutes"] = (
    df["next_timestamp"] - df["timestamp"]
).dt.total_seconds() / 60


transitions = df.dropna(
    subset=["next_activity", "next_timestamp"]
).copy()


# Keep valid non-negative elapsed intervals
transitions = transitions[transitions["elapsed_minutes"] >= 0].copy()
```

Appendix D: Repository Inventory

Table D.1: Contents of the four submitted branches

Branch	Contents
main	README.md (consolidated project documentation, architecture, findings, recommendations, replication steps)
akshar	careflow_impute_scale.py; careflow_cases_imputed / standardized / normalized.csv; careflow_events_scaled.csv; careflow_model_ready_scaled.csv; imputation_benchmark.csv; scaling_checks.csv; scaler_parameters.json; branch README; project assignment letter (PDF)
sagar	14 notebooks (day3 to day16); data/processed_old (4 baseline files) and data/processed_new (14 files incl. audit, outlier, segmentation, peak-load, feature-significance and final-metrics summaries); four Power BI files; STAKEHOLDER_SUMMARY.md; day18_notes.md; dashboard_p4_notes.md; README.md
prachi	18 Python scripts in src; 19 processed summary CSVs in data/processed; 16 documents plus three PNG charts and two SVG process maps in docs; sample_event_log.csv; README.md

Appendix E: Glossary

Table E.1: Terms and abbreviations

Term	Meaning
Acuity / ESI	Emergency Severity Index, a five-level triage score from 1 (most urgent) to 5 (least urgent)
Activity	A named step recorded in the event log, for example Triage or Lab Result
Boarding	A patient remaining in the department for an extended time, here approximated by a visit that crosses midnight
Case	One emergency department visit, identified by a case_id
Chronological split	Division of the data into earlier (training) and later (test) cases
DFG	Directly-follows graph: nodes are activities and edges show which activity directly follows which
Door-to-doctor time	Time from registration to the first physician assessment
Event log	A table of timestamped events, each belonging to a case and an activity
IQR	Interquartile range: the difference between the third and first quartiles
LOS	Length of stay: time from registration to discharge
Loopback	A return to an earlier step of the pathway, here repeat triage, documentation review and doctor reassessment
Min-max normalisation	Rescaling of a feature to the interval from 0 to 1 using the training minimum and maximum
NRMSE	Normalised root mean squared error, the RMSE divided by the standard deviation of the true values
One-hot encoding	Representation of a category by a set of 0/1 columns, one per category
PM4Py	An open-source Python library for process mining
Process mining	Discovery and analysis of processes from event logs
Standardisation (z-score)	Rescaling of a feature to zero mean and unit variance using the training mean and standard deviation
Structural missingness	A value that is absent because the event it describes did not occur
Trace	The ordered sequence of activities of one case
Variant	A distinct trace; the 1,000 visits form six variants

Appendix F: Extended Event-Log Excerpt

The table below shows the first 40 rows of the event table, covering the first several visits in chronological order of case identifier. The column is_loopback marks events on the rework path.

Table F.1: First 40 rows of the event table

Case	Activity	Timestamp	Order	Min since prev	Physician	ESI	Loopback
CF00001	Registration	2026-01-06 15:31	1	0	DR06	3	0
CF00001	Triage	2026-01-06 15:53	2	22	DR06	3	0
CF00001	Doctor Assessment	2026-01-06 16:13	3	20	DR06	3	0
CF00001	Discharge	2026-01-06 16:19	4	6	DR06	3	0
CF00002	Registration	2026-02-13 08:10	1	0	DR11	2	0
CF00002	Triage	2026-02-13 08:14	2	4	DR11	2	0
CF00002	Doctor Assessment	2026-02-13 08:20	3	6	DR11	2	0
CF00002	Discharge	2026-02-13 08:44	4	24	DR11	2	0
CF00003	Registration	2026-02-03 00:25	1	0	DR02	3	0
CF00003	Triage	2026-02-03 00:41	2	16	DR02	3	0
CF00003	Doctor Assessment	2026-02-03 01:34	3	53	DR02	3	0
CF00003	Discharge	2026-02-03 01:56	4	22	DR02	3	0
CF00004	Registration	2026-02-12 07:45	1	0	DR06	5	0
CF00004	Triage	2026-02-12 08:08	2	23	DR06	5	0
CF00004	Doctor Assessment	2026-02-12 08:56	3	48	DR06	5	0
CF00004	X-Ray	2026-02-12 09:05	4	9	DR06	5	0
CF00004	Radiology Result	2026-02-12 09:29	5	24	DR06	5	0
CF00004	Documentation Review	2026-02-12 09:32	6	3	DR06	5	0
CF00004	Lab Collection	2026-02-12 09:48	7	16	DR06	5	0
CF00004	Lab Result	2026-02-12 10:27	8	39	DR06	5	0
CF00004	Discharge	2026-02-12 10:51	9	24	DR06	5	0
CF00005	Registration	2026-02-14 23:27	1	0	DR12	3	0
CF00005	Triage	2026-02-14 23:32	2	5	DR12	3	0
CF00005	Doctor Assessment	2026-02-15 00:05	3	33	DR12	3	0
CF00005	X-Ray	2026-02-15 00:24	4	19	DR12	3	0
CF00005	Radiology Result	2026-02-15 00:46	5	22	DR12	3	0
CF00005	Triage	2026-02-15 00:53	6	7	DR12	3	1
CF00005	Documentation Review	2026-02-15 01:02	7	9	DR12	3	1
CF00005	Doctor Reassessment	2026-02-15 01:15	8	13	DR12	3	1
CF00005	Lab Collection	2026-02-15 01:30	9	15	DR12	3	0
CF00005	Lab Result	2026-02-15 02:18	10	48	DR12	3	0
CF00005	Discharge	2026-02-15 02:32	11	14	DR12	3	0
CF00006	Registration	2026-01-10 22:23	1	0	DR02	4	0
CF00006	Triage	2026-01-10 22:33	2	10	DR02	4	0
CF00006	Doctor Assessment	2026-01-10 23:05	3	32	DR02	4	0

Case	Activity	Timestamp	Order	Min since prev	Physician	ESI	Loopback
CF00006	X-Ray	2026-01-10 23:26	4	21	DR02	4	0
CF00006	Radiology Result	2026-01-10 23:42	5	16	DR02	4	0
CF00006	Documentation Review	2026-01-10 23:50	6	8	DR02	4	0
CF00006	Lab Collection	2026-01-11 00:05	7	15	DR02	4	0
CF00006	Lab Result	2026-01-11 00:25	8	20	DR02	4	0

Appendix G: Feature Store Column Reference

The table below documents every column of careflow_model_ready_scaled.csv (41 columns). Columns whose names begin with target_ are prediction targets and must not be used as inputs; the identifier columns are keys only.

Table G.1: Columns of the model-ready feature store

Column(s)	Type	Description
case_id	Identifier	Visit identifier
case_start	Timestamp	Registration time, used for the chronological split
split	Label	train (800 cases) or test (200 cases)
age	Numeric	Age in years
age_z, age_mm	Numeric	Age standardised (training mean 55.0) and min-max normalised
age_group_18-39, age_group_40-59, age_group_60+	Binary	One-hot age band
acuity	Ordinal	ESI level from 1 to 5
acuity_z, acuity_mm	Numeric	Acuity standardised and min-max normalised
start_hour_sin, start_hour_cos	Numeric	Cyclical coordinates of the arrival hour
is_weekend	Binary	1 for Saturday and Sunday
start_weekday_Monday … start_weekday_Sunday (7 columns)	Binary	One-hot weekday of arrival
shift_Day, shift_Evening, shift_Night	Binary	One-hot shift of arrival
resource_DR01 … resource_DR12 (12 columns)	Binary	One-hot attending physician
target_total_duration_min	Target	Length of stay in minutes
target_total_duration_min_z	Target	Length of stay standardised with the training mean and standard deviation
target_loopback	Target	1 if the visit included a loopback
target_ends_next_day	Target	1 if the visit crosses midnight

Appendix H: Additional Cross-Tabulations

The following tables give the case counts behind several percentages quoted in the report. They allow readers to recompute shares or to apply their own tests.

Table H.1: Cases by acuity level and pathway variant

ESI	V1	V2	V3	V4	V5	V6	Total
1	5	4	5	3	6	3	26
2	45	23	24	28	14	10	144
3	116	104	63	55	53	38	429
4	76	68	47	48	43	27	309
5	25	17	10	12	17	11	92

Table H.2: Cases by shift and pathway variant

Shift	V1	V2	V3	V4	V5	V6	Total
Day	79	61	47	48	45	32	312
Evening	102	79	57	47	43	30	358
Night	86	76	45	51	45	27	330

Table H.3: Cases by attending physician and pathway variant

Physician	V1	V2	V3	V4	V5	V6	Total	Loopback cases (V4 + V6)
DR01	32	24	15	17	7	7	102	24
DR02	23	16	12	13	13	9	86	22
DR03	24	15	12	11	8	9	79	20
DR04	21	14	18	10	14	5	82	15
DR05	21	32	10	6	14	6	89	12
DR06	20	15	8	16	8	8	75	24
DR07	35	19	20	12	10	10	106	22
DR08	17	15	15	9	8	10	74	19
DR09	23	17	9	12	16	8	85	20
DR10	16	15	7	16	7	3	64	19
DR11	19	20	11	5	14	4	73	9
DR12	16	14	12	19	14	10	85	29